
HIGHER EDUCATION COMMISSION OF PAKISTAN

AI, DATA SCIENCE AND NETWORKING

One Course Cluster, Three Curriculum Editions

2017 • 2023 • 2025

A version-by-version comparison, a pathway, five semester schemes for BSIT and BSCS (IT specialization), a CCNA v2.0 alignment, and the published contents of every course



Every claim about the curriculum is taken from the three source booklets and cited to them; Part 7 additionally cites Cisco.

Scope. This is not a review of the three curricula. It covers one group of courses only — artificial intelligence, machine learning, data mining, data science, deep learning, natural language processing and the courses that sit alongside them — together with the infrastructure and networking courses that the group runs on and competes with for credit: *Computer Networks*, *System & Network Administration*, *Cloud Computing* and the ten networking courses HEC 2025 adds. Part 7 maps those against Cisco's CCNA 200-301 v2.0 blueprint. Everything else in the three booklets (software engineering, cyber security, graphics, the general education block) appears only where it is a prerequisite, a competitor for elective credit, or a source of overlap.

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Summary

What the three editions did to this group of courses

1. **In 2017 a BSIT student could not study machine learning at all.** Artificial Intelligence was not in the BSIT core — it was a BSCS course — and no undergraduate course called Machine Learning, Deep Learning, Data Mining or Data Science existed anywhere in the booklet. The cluster was a postgraduate subject.
2. **In 2023 it came down to the bachelor level completely**, far enough to support two whole degrees, BSAI and BSDS. Artificial Intelligence became compulsory for every computing degree in Semester 3. This is the change that makes an undergraduate data pathway possible at all.
3. **In 2025 it became a named 24-credit specialization** that any BS Computer Science student may elect, and gained Generative AI, Agentic AI, MLOps and Data Engineering. But the separate BSIT degree disappears: Information Technology survives only as one specialization among fourteen.

The four findings that change what a department should do

1. **HEC 2023 specifies the AI, Data Science and Cyber Security clusters in full, and the Information Technology cluster not at all.** Forty-six of roughly 180 courses have published contents. A department teaching System & Network Administration or Cloud Computing under 2023 has a title, a credit count, and nothing else — the 2017 outlines are the only detailed text HEC has ever published for them.
2. **HEC 2025 publishes no prerequisite and no outline for any of its 215 specialization courses.** Sequencing stops being a curricular rule and becomes a departmental responsibility. The prerequisite chains in Part 6 are the ones to adopt, because they are the only ones anybody published.
3. **Probability & Statistics stops being compulsory in 2025.** It is item four in an optional pool, and the free slots fall in Semester V. A student who does not choose it reaches Machine Learning with no statistics at all. This is the single largest risk to a data track.
4. **Advanced Computer Networks and Routing & Switching exist in no edition, under any name.** Networking depth arrives only in 2025, as a twenty-course cluster in which *Internetworking* is the functional routing-and-switching course.

If you are aiming at CCNA

Students entering from Fall 2026 will all sit **CCNA 200-301 v2.0**, which replaces v1.1 on 3 February 2027. Of its twenty-nine published topics, eight begin *Troubleshoot* or *Diagnose* and eight begin *Configure*: **55% of the exam is hands-on work at a command line**, against 21% descriptive. Under v1.1 the word “troubleshoot” appeared in no topic statement at all.

No HEC networking outline in any edition names a VLAN, a trunk, a spanning tree, OSPF, an access control list, or a command line. Domains 2.0 and 3.0 together are 45% of the exam and one course in one edition touches either. The gap is practice, not knowledge, and it is a laboratory decision rather than a syllabus one.

✓ What to do first

1. **Adopt Scheme B** (Part 8) for a data track, or **Scheme E** for a networking and CCNA track. Both spend only elective credit, so neither needs HEC approval of a new programme.

2. **Write the 2023 prerequisite chains into your own scheme of studies** and encode them in the student information system. Under 2025 nothing else will enforce them.
3. **Mandate Probability & Statistics as IDS-III** for anyone on a data or AI specialization under 2025.
4. **Restore the laboratory hour.** 2025 lists these courses as 3+0 or (2+1)/3; offer them as 2+1. Neither this cluster nor CCNA can be taught from a whiteboard.
5. **Write “configure” and “diagnose” into your local course learning outcomes** instead of “explain”. Because 2025 publishes no outlines you are free to, and it is the cheapest change available.

How to read the rest

If you are...	Read
Setting a scheme of studies	Part 8 for the five schemes, then Part 9 for what goes wrong
Writing a course file	Part 10, which reproduces every published outline, prerequisite, outcome and reading list verbatim
Deciding whether to run a CCNA track	Part 7, then Scheme E in Part 8
Recognising a course across editions	Part 2 for the status matrix, Part 3 for what each edition says about it
Advising a student on course order	Part 6, the pathway diagram
Working out which courses duplicate each other	Part 5, the overlap map

PART 1

The Three Sources, and What Each One Is

The three documents are not three drafts of the same thing. They differ in what they cover, and that difference explains most of what follows.

	2017	2023	2025
Full title	BS & MS (CS, SE & IT) Curriculum	BS Curriculum, Computing Disciplines	HEC Curriculum of Computer Science
Covers	BSCS, BSSE, BSIT, and the MS programmes	Eleven BS degrees, including BSAI, BSDS and BSIT, plus an associate degree	BS Computer Science only, plus an associate degree in computing
Length	171 pages	100 pages	86 pages
Degree size	130 credit hours	130 credit hours	130 credit hours

	2017	2023	2025
Is there a BSIT?	Yes, a full separate programme	Yes, a full separate programme	No. IT survives only as one of fourteen specializations inside BSCS
How courses are specified	Credit hours, prerequisites, CLOs with Bloom levels, a full topic outline, reference books	The same, plus a course introduction paragraph	CLOs with Bloom levels only. No topic outline, no prerequisites, no reference books
How the cluster is organised	A flat list of 75 bachelor courses; universities pick electives from it	Each degree gets a named domain core and a domain elective list	Fourteen named specialization clusters of 15–21 courses each

💡 The single most important structural fact

In **2017** the AI and data science cluster is essentially a *postgraduate* subject. Machine learning, deep learning, data mining and data science do not exist as undergraduate courses at all — they appear only inside MS Data Science. By **2023** the cluster has moved down to the bachelor level so completely that it supports two whole degrees, BSAI and BSDS. By **2025** it has become a named 24-credit specialization that any BS Computer Science student may elect. Three editions, three completely different answers to “when should a student meet machine learning?”

1.1 Where the elective credit actually is

For a BSIT or BSCS student the cluster is almost never compulsory. It is bought with elective credit, so the size and placement of the elective block decides how much of it a student can take.

	2017 (BSIT)	2023 (BSIT)	2025 (BSCS)
Elective budget	15 cr, 5 courses (“IT Elective 1–5”)	21 cr, 7 courses (Domain Electives)	24 cr, 8 courses (Elective I–VIII)
Where they fall	Semesters 5–8	Semester 5 (×2), Semester 6 (×4), Semester 7 (×1)	Semester VI (×4), Semester VII (×4)
Cluster courses available to spend it on	Five: AI, Big Data Analytics, Computer Vision, Digital Image Processing, NLP	All of them — the master list of BS courses carries every course used by BSAI and BSDS	Fifteen in the Data Science cluster, fifteen in the AI cluster
Constraint	No ML, DL, DM or DS course exists to elect	The BSIT elective examples name none of the cluster; a department must reach into the master list deliberately	Choosing the named <i>Information Technology</i> specialization spends all eight electives on courses with no AI or data content

PART 2

Where Each Course Lives in Each Edition

This is the whole comparison on one page. Read a row to see what happened to a course over eight years; read a column to see what a student in that edition could actually take.

✔ Compulsory 🟡 Available as an elective 🎓 Postgraduate only ✖ Does not exist

Course	2017	2023	2025	Note
Artificial Intelligence	🟡	✔	✔	Compulsory for <i>every</i> computing degree from 2023. In 2017 it was a BSCS domain core course and entirely absent from BSIT
Machine Learning	🎓	🟡	🟡	The pivotal change. MS-only in 2017; a BSAI domain core and BSDS elective in 2023; in both the DS and AI clusters in 2025
Deep Learning	🎓	🟡	🟡	2023 splits it in two — <i>ANN & Deep Learning</i> and <i>Deep Learning</i> ; 2025 merges them back into one course
Data Mining	✖	🟡	🟡	Absent from 2017 at every level. A BSDS domain core in 2023
Introduction to Data Science	🎓	🟡	🟡	2017 has <i>Tools and Techniques in Data Science</i> at MS level only
Natural Language Processing	🟡	🟡	🟡	Present in all three, but the content is unrecognisable between 2017 and 2025
Big Data Analytics	🟡	🟡	🟡	The most stable course in the cluster
Computer Vision	🟡	🟡	🟡	A BSAI domain core in 2023
Data Visualization	🎓	🟡	🟡	A BSDS domain core in 2023
Data Warehousing & BI	✖	🟡	🟡	Renamed to plain <i>Business Intelligence</i> in 2025, with warehousing moved into <i>Data Engineering</i>
Reinforcement Learning	✖	🟡	🟡	2017 covers it as a topic inside AI, never as a course
Knowledge Representation & Reasoning	✖	🟡	🟡	A BSAI domain core in 2023
Programming for AI	✖	🟡	🟡	A BSAI domain core in 2023
Speech Processing	✖	🟡	🟡	
Text Mining	✖	🟡	✖	Dropped in 2025; its ground is split between NLP and Information Retrieval

Course	2017	2023	2025	Note
Advanced / Computational Statistics	✗	🟡	🟡	Renamed <i>Computational Statistics</i> in 2025
Generative AI	✗	✗	🟡	New in 2025, and in both the DS and AI clusters
Agentic AI	✗	✗	🟡	New in 2025
MLOps	✗	✗	🟡	New in 2025
Data Engineering	✗	✗	🟡	New in 2025
Information Retrieval	✗	✗	🟡	New as a course; 2017 taught it inside NLP
Data Ethics & Security	✗	✗	🟡	New in 2025
Optimization Techniques	✗	✗	🟡	New in 2025
Stochastic Processes	🟡	✗	🟡	In the 2017 bachelor list, gone in 2023, back in the 2025 AI cluster
Digital Image Processing	🟡	🟡	✗	Absorbed into Computer Vision by 2025

The infrastructure the cluster runs on

Computer Networks	✓	✓	✓	Compulsory in every degree in every edition. The only course in this document that is <i>never</i> an elective
System & Network Administration	✓	✓	🟡	BSIT-compulsory in 2017 and 2023; demoted to one elective among fifteen in the 2025 IT specialization
Cloud Computing	✗	🟡	✓	The mirror image: named but never specified in 2017, an unspecified BSIT elective in 2023, a compulsory major course for every BSCS student in 2025
Virtual Systems & Services	🟡	🟡	✗	2017's nearest thing to a cloud course — a hypervisor course. Gone by 2025, its ground taken by Cloud Computing
IT Infrastructure	✓	✓	🟡	The same trajectory as System & Network Administration
Network Security	✗	🟡	🟡	In three 2025 clusters at once: Cyber Security, Information Technology, and Network Infrastructure & Cloud Computing

Networking depth — all of it new in 2025 (Part 7)

Course	2017	2023	2025	Note
Internetworking	✗	✗	👤	Routers, switches and gateways. The nearest thing in any edition to a <i>Routing & Switching</i> course
Data Communication	✗	✗	👤	The physical and link layer, split out of Computer Networks
Network Management	✗	✗	👤	
Network Performance	✗	✗	👤	The only course in any edition that names QoS
Wireless and Mobile Networks	✗	✗	👤	2023 has <i>Wireless and Mobile Security</i> , which is not the same course
Software Defined Networks	✗	✗	👤	Controller-based networking, and the entry point to CCNA domain 6
Network Programmability & Automation	✗	✗	👤	APIs, scripts and playbooks
Network Automation & Orchestration	✗	✗	👤	
Next-Generation Networks & Edge Computing	✗	✗	👤	5G and edge
Enterprise Network Infrastructure Engineering Design	✗	✗	👤	The design capstone of the 2025 infrastructure cluster
<i>Advanced Computer Networks</i>	✗	✗	✗	Exists in no edition under that name , at either level
<i>Routing & Switching</i>	✗	✗	✗	Exists in no edition under that name. <i>Internetworking</i> is the functional equivalent

💡 Why three networking courses belong in a document about machine learning

Three reasons, and none of them is padding. **They are the platform.** Big Data Analytics is a Spark course, MLOps is a deployment course and Data Engineering is a pipeline course — all three assume the machine underneath. **They are the competition.** A BSIT student has seven elective slots; every one spent on Cloud Computing is one not spent on Deep Learning, and Part 8 prices that trade-off explicitly. **They are where HEC 2025 quietly put the bridge:** the *Network Infrastructure & Cloud Computing* specialization contains both *Machine Learning* and *MLOps*, which makes it the only 2025 cluster outside Data Science and AI to reach into this material at all.

⚠️ A BSIT student in 2017 could not study machine learning

This is worth stating plainly because it explains why so many Pakistani IT departments still have no ML teaching capacity. Under the 2017 curriculum the BSIT computing core was ten courses — programming, data structures, discrete structures, operating systems, databases, software engineering, networks, information security and the project — and *artificial intelligence was not among them*. It was a BSCS domain core course only. The 75-course bachelor list offered AI, Big Data Analytics, Computer Vision, Digital Image Processing and NLP; it contained no

course called Machine Learning, Deep Learning, Data Mining or Data Science. A department that wanted to teach machine learning had to invent the course itself.

PART 3

Course by Course, Version by Version

Each card below takes one course and shows what each edition says about it. Credit-hour notation follows the source: 2017 writes 3-1 or 3(2,1), 2023 writes 3 (2-3) meaning two lecture hours and three laboratory hours per week, and 2025 writes 3+0 or 2+1.

These cards are the *analysis*. The full published text — every topic outline, prerequisite, learning outcome and reading list, verbatim, for forty-three courses across all three editions — is reproduced in Part 10, which is the part to open when writing an actual course file.

3.1 The AI and data science courses

📖 Artificial Intelligence — *the hinge the whole cluster turns on*

2017 BSCS domain core, 3+1. Absent from BSIT.

Prerequisite *Data Structures and Algorithms*. Outline: basic components and branches of AI; reasoning and knowledge representation, propositional and first-order logic; uninformed, informed and local search; constraint satisfaction; adversarial search with minimax and alpha-beta; learning (supervised, unsupervised, reinforcement); uncertainty and fuzzy logic; recent trends. Russell & Norvig.

2023 Computing core for every degree, Semester 3, 3 (2-3).

Prerequisite *Object Oriented Programming* — notably weaker than 2017's data-structures requirement. Outline drops the formal logic emphasis and adds case studies (General Problem Solver, Eliza, Student, Macsyma), ANN and NLP as topics, and mandates Python.

2025 Major core, Semester V, 3 credit hours.

No prerequisite published and no topic outline — only CLOs. The course now sits two years later in the degree than it did in 2023.

➔ **What changed:** It became compulsory for everyone, then slid from Semester 3 to Semester V. The 2023 move is the single most consequential change in these three documents for IT students; the 2025 move squeezes everything that depends on it into the final two years.

📖 Machine Learning — *the course that did not exist for undergraduates in 2017*

2017 MS Data Science programme core only. 3 credit hours.

One of three MS(DS) core courses, alongside *Statistical and Mathematical Methods for Data Science* and *Tools and Techniques in Data Science*. There is no undergraduate equivalent anywhere in the booklet.

2023 BSAI domain core (Semester 4); BSDS domain elective. 3 (2-3).

Prerequisite *Artificial Intelligence*. Five CLOs from C1 to C6. Outline is the classical Mitchell syllabus: concept learning and version spaces; decision trees, naive Bayes, ANNs, SVMs; overfitting, pruning, accuracy measurement; linear and logistic regression; hierarchical and *k*-means clustering, SOM, *k*-NN; semi-supervised EM; reinforcement learning with HMMs, Monte Carlo and MDPs; ensembles, bagging and boosting.

2025 In three clusters at once: Data Science, AI, and Software Engineering.

3+0 in the DS and AI clusters, 2+1 in the Software Engineering cluster — the same course with different laboratory provision depending on which list you read it from. CLOs add an explicit ethics outcome (A3: Valuing).

➔ **What changed:** From postgraduate to ubiquitous in eight years. The 2023 outline is the most detailed specification of the three and remains the best reference for what the course should contain; 2025 gives it an ethics outcome but no syllabus.

Deep Learning — *one course, then two, then one again***2017 MS Data Science specialization core. 3 credit hours.**

One of four courses from which MS(DS) students choose two. *Deep Reinforcement Learning* also appears in the MS elective list. Nothing at bachelor level.

2023 Two separate courses.

Artificial Neural Networks & Deep Learning, a BSAI domain core, prerequisite *Programming for AI* — covers perceptron and Adaline, Hebbian learning, Kohonen maps, associative and bidirectional associative memory, Boltzmann machines, backpropagation, CNNs, autoencoders, deep belief networks. Then *Deep Learning*, an AI elective with the first course as its prerequisite — CNNs and RNNs, BPTT and LSTM, GANs, sparse coding, dropout and batch normalisation, ResNet and GoogLeNet, CuDNN.

2025 A single Deep Learning course in both the DS and AI clusters. 3+0.

Four CLOs in the DS cluster, three in the AI cluster. No outline.

➔ **What changed:** 2023's split is genuinely awkward — the two outlines overlap on generalization theory, multi-layer perceptrons, backpropagation, CNNs and autoencoders, so a student taking both repeats a third of the material. 2025 merging them is the right call, but it also means the historical neural-network material (Hebbian learning, Boltzmann machines, SOMs) quietly disappears.

Data Mining — *absent, then a data science pillar***2017 Does not exist, at either level.**

No course of this name appears in the 75-course bachelor list or in any MS programme. Association rules and frequent patterns are taught nowhere.

2023 BSDS domain core (Semester 5); BSAI domain elective. 3 (2-3).

Prerequisites *Advanced Statistics* and *Introduction to Data Science*. Outline: preprocessing and summary statistics; association rule mining with Apriori and FP-trees; supervised classification (decision trees, naive Bayes, k -NN, SVM); unsupervised classification (k -means, k -median, hierarchical and divisive clustering, Kohonen maps); outlier and anomaly detection; web and social network mining. Python.

2025 In both the DS and AI clusters. 3+0.

Three CLOs, pitched at knowledge discovery and decision making rather than algorithms.

➔ **What changed:** It appeared from nothing in 2023 and stayed. The overlap with Machine Learning is severe and is discussed in Part 5 — roughly half the 2023 Data Mining outline is also in the 2023 Machine Learning outline.

Introduction to Data Science — *the gateway course***2017 MS only, as Tools and Techniques in Data Science, 2+1.**

An MS(DS) core course. The undergraduate programmes have nothing comparable.

2023 BSDS domain core, Semester 4. 3 (2-3).

Prerequisite *Artificial Intelligence*. Outline: what data science is, datafication, skill sets; statistical inference, populations and samples, fitting a model, Python; exploratory data analysis and the data science process; basic algorithms (linear regression, k -NN, k -means, naive Bayes); feature generation and selection; dimensionality reduction with SVD and PCA; mining social network graphs; data visualization; privacy, security and ethics.

2025 Data Science cluster, 3+0.

Three CLOs: understand the principles, apply EDA and the data science process, apply basic ML algorithms to real problems of moderate complexity.

➔ **What changed:** Stable in intent across 2023 and 2025 — it is the process-and-EDA course, with a taste of modelling. Its prerequisite on *Artificial Intelligence* in 2023 is odd, since almost nothing in the outline depends on search or knowledge representation; the real dependencies are programming, statistics and databases.

📖 Natural Language Processing — the same name, three different courses

2017 Bachelor elective, 3(3,0), no prerequisite.

Purely symbolic and statistical: deterministic and stochastic grammars, parsing algorithms, CFGs, semantics and semantic roles, corpus methods, n -grams and HMMs, smoothing and backoff, POS tagging and morphology, information retrieval, the vector space model, precision and recall, information extraction, machine translation, bag of words. No neural content whatsoever.

2023 BSAI domain elective; BSDS-adjacent. 3 (2-3).

Prerequisite *Artificial Neural Networks & Deep Learning* — so the course is now explicitly downstream of deep learning. The CLOs still read like the 2017 ones.

2025 In both the DS and AI clusters, 3+0.

DS-cluster CLOs name tokenization, POS tagging, parsing, NLTK, sentiment analysis and text classification. AI-cluster CLOs are broader and add an ethics dimension.

➔ **What changed:** The syllabus of record is still recognisably the 2000s course. The one real change is the 2023 prerequisite, which correctly places NLP after deep learning — but the published content never caught up. Neither 2023 nor 2025 mentions transformers, attention, embeddings or language models anywhere in the NLP specification.

📖 Big Data Analytics — the most stable course in the cluster

2017 Bachelor elective, 3(2,1).

Prerequisites *Probability and Statistics* and *Programming Fundamentals*. Outline: big data platforms, Hadoop, storage and analytics, ML algorithms on big data, recommendation, clustering and classification, graph computing and graph analytics, graphical models and Bayesian networks, visualization, cognitive mobile analytics. Leskovec, Rajaraman & Ullman; Tom White.

2023 BSDS domain elective. 3 (2-3).

Prerequisite *Introduction to Data Science*. Outline moves decisively to Spark: HDFS, MapReduce, YARN, Scala basics and advanced, RDDs, Spark SQL, ML on Hadoop and Spark, Spark Streaming.

2025 DS cluster, 3+0.

Three CLOs, still naming Hadoop, MapReduce and Apache Spark explicitly — the only 2025 course in the cluster that names specific technologies in its outcomes.

➔ **What changed:** Present in all three editions with a coherent story: Hadoop-centric in 2017, Spark-centric from 2023. In 2025 it now shares ground with the new *Data Engineering* course and the two need deliberate separation.

The 2025 arrivals — *Generative AI, Agentic AI, MLOps, Data Engineering*

2017 None of these exist.

Not as courses, not as topics.

2023 None of these exist.

The word “generative” does not appear in the 2023 course specifications for this cluster.

2025 Four new courses, all 3+0.

Generative AI (in both the DS and AI clusters): GANs, VAEs and large language models; implementing and fine-tuning with TensorFlow or PyTorch; ethical and societal implications. **Agentic AI** (AI cluster): autonomous agents that reason, plan and use tools; integration with APIs, databases and external tools; responsible design. **MLOps** (AI cluster): end-to-end pipelines for deploying, monitoring and maintaining models; modern tooling; ethical and security assessment. **Data Engineering** (DS cluster): pipelines, ETL, storage systems, data integration and transformation, architectures for big data and real-time analytics.

➔ **What changed:** This is the clearest signal of intent in the 2025 booklet. It is also the hardest part to staff: these four courses have the least settled syllabi, the fastest-moving tooling, and no published outline to anchor them.

3.2 The infrastructure courses

Computer Networks — *the most stable course in all three booklets*

2017 Computing core for every degree, 3+1. No prerequisite.

BSIT Semester IV. Outline: protocol architecture, topologies, layered architecture, physical and data link layer, multiple access, circuit and packet switching, LAN technologies, wireless, MAC addressing, network devices, IPv4 and IPv6, addressing, subnetting, CIDR, routing protocols, transport layer, ports and sockets, flow and congestion control, application layer protocols.

2023 Computing core for every degree, 3 (2-3). No prerequisite.

BSIT Semester 3 — one semester earlier than 2017. The published outline is **word for word identical to the 2017 outline**, down to the phrase “latest trends in computer networks”. Only the laboratory allocation changed.

2025 Major core, Semester V, 3 credit hours.

CLOs only. Two of the five are new in substance: *analyze network security threats and mitigation techniques* and *design simple network architectures to meet organisational needs* — the course is asked to reach past protocol description into design.

➔ **What changed:** Almost nothing. It is compulsory in every degree of every edition, it has never had a prerequisite, and its 2023 syllabus is a verbatim copy of its 2017 syllabus. If you want a control against which to measure how much the rest of this document moved, this is it.

System & Network Administration — *compulsory, then compulsory, then optional*

2017 BS-IT core (compulsory), 3 (3,0). Prerequisite *Operating Systems*.

BSIT Semester V. A detailed and frankly practical outline: system administration components, Microsoft and Linux server environments, server hardware costing, maintenance contracts and spare parts, data integrity, client-server configuration, remote console access, comparative analysis of operating systems, Linux installation and verification, users and groups, software management, network services and monitoring tools, boot and process management, IP tables and filtering, securing network traffic, advanced file systems and logs, Bash shell scripting.

2023 BSIT domain core, Semester 5, 3 (2-3).

No course contents are published. The 2023 booklet specifies only the twenty-nine courses common to all computing degrees plus the AI, Data Science and Cyber Security clusters — there is no Information Technology table at all. A department teaching this course under HEC 2023 has a title, a credit count and nothing else.

2025 One elective among fifteen in the Information Technology specialization. 2+1.

CLOs only. Notably, two of the five are not technical: *demonstrate professionalism in managing shared computing resources* and *work effectively in teams to maintain IT infrastructure*.

➔ **What changed:** It went from a compulsory course with a two-hundred-word syllabus to an optional course with five outcome statements. For a BSIT department this is the sharpest loss of specification in the three editions, and the 2017 outline remains the only usable description of the course that HEC has ever published.

Cloud Computing — the exact opposite trajectory**2017 Named in the BS-SE elective list at 3-0, and nowhere else.**

It is **not** in the seventy-five-course bachelor list and has **no course specification** anywhere in the booklet. A BSIT student's nearest equivalent was *Virtual Systems and Services* (3 cr, prerequisite Programming Fundamentals) — a hypervisor course covering VMware, Xen and Hyper-V, para-virtualised components, and processor support for virtualisation. That is virtualisation, not cloud: no service models, no elasticity, no billing, no managed services.

2023 BSIT domain elective example, 3 (2-3), and in the master course list.

Also **unspecified** — same gap as System & Network Administration.

2025 Major core course, Semester VI, 3 credit hours. Compulsory for every BS Computer Science student.

CLOs: describe IaaS, PaaS and SaaS; deploy and manage applications in cloud environments; analyze benefits and challenges; implement basic cloud security and compliance; evaluate cloud solutions for scalability, cost and performance. It also anchors an entire specialization — *Network Infrastructure & Cloud Computing*, twenty courses including DevOps, DevSecOps, Infrastructure as Code, Microservices and Docker, Software Defined Networks, Cloud-Native Application Development, and Next-Generation Networks and Edge Computing.

➔ **What changed:** From a name with no syllabus to a compulsory course plus a twenty-course specialization in eight years — the steepest promotion in the three editions. And that specialization is where HEC 2025 put *Machine Learning* and *MLOps*, making it the only cluster outside Data Science and AI that reaches into this material.

i The specification asymmetry in HEC 2023

It is worth naming what the three cards above expose. HEC 2023 publishes full course contents — introduction, CLOs with Bloom levels, prerequisites, topic outline and reference books — for the Artificial Intelligence cluster, the Data Science cluster and the Cyber Security cluster. It publishes *nothing* for the Information Technology cluster. Web Technologies, Database Administration & Management, System & Network Administration, Information Technology Infrastructure, Virtual Systems & Services, Enterprise Systems, Web Engineering and Cloud Computing all appear in the BSIT tables and in the master course list, and none of them has a specification. Computer Networks and Network Security are specified only because they are common to every computing degree. The practical consequence is that a BSIT department writing course files for its own domain core is on its own, and should use the 2017 outlines — which are detailed, still broadly accurate, and the only ones that exist.

PART 4

What Stays the Same, and What Changes

4.1 The invariants

Underneath the churn there is a stable spine. A department planning a pathway can rely on all of the following holding across every edition.

1. **The degree is 130 credit hours** in all three editions, over eight semesters, with a final year project.
2. **The programming and data foundation never changes.** Programming Fundamentals, Object Oriented Programming, Data Structures and Database Systems are compulsory in every edition of every computing degree, and they always come in the first three or four semesters.
3. **Artificial Intelligence is the gateway.** Where the cluster is taught at all, AI comes first — as a stated prerequisite in 2023, as the earliest compulsory cluster course in 2025.
4. **The cluster is bought with elective credit.** No edition makes machine learning compulsory for an IT student. It is always an elective choice, which means it is always a departmental decision rather than an HEC one.
5. **Three courses appear in all three editions:** Natural Language Processing, Big Data Analytics and Computer Vision. If a department has taught anything from this cluster since 2017, it is most likely one of these.
6. **Computer Networks is the one true constant.** It is compulsory in every degree of every edition, it has never carried a prerequisite, and its 2023 syllabus is a verbatim reproduction of its 2017 syllabus. Every other course in this document moved.
7. **Mathematics support is consistent in intent:** calculus, linear algebra and probability and statistics. What changes is whether probability is compulsory — see the warning in Part 9.
8. **Bloom's taxonomy labelling of course learning outcomes** is used in all three, so outcomes remain comparable even when outlines are not.

4.2 2017 to 2023: the cluster comes down to the bachelor level

- **Artificial Intelligence becomes compulsory for every computing degree**, BSIT included, in Semester 3. This is the change that makes an undergraduate data science pathway possible at all.
- **Eight courses appear from nothing:** Machine Learning, Data Mining, Introduction to Data Science, Data Visualization, Data Warehousing & Business Intelligence, Reinforcement Learning, Knowledge Representation & Reasoning and Programming for AI all become undergraduate courses with published outlines.
- **Two new degrees**, BSAI and BSDS, give the cluster a home and, more usefully for everyone else, give it a specified curriculum that other degrees can borrow from.
- **Prerequisite chains are published and enforced.** 2023 is the only edition that states, course by course, what must come first. Those chains are the backbone of the pathway in Part 6.
- **Laboratory hours are made systematic:** nearly every cluster course becomes 3 (2-3) — two hours of lecture and three of laboratory.
- **But the Information Technology cluster is left unspecified.** 2023 writes full course contents for the AI, Data Science and Cyber Security clusters and none at all for the IT one — System & Network Administration, Cloud Computing, IT Infrastructure, Web Technologies, Database Administration and the rest have a title and a credit count and nothing more.

4.3 2023 to 2025: from degrees to specializations

- **The separate degrees go away** — at least in this booklet, which covers Computer Science only. Data Science and Artificial Intelligence become two of fourteen specializations, each defined by a cluster of 15 elective courses from which a student takes seven or eight.
- **The specialization may be named on the degree or the transcript**, or not at all: a university may run a generic BSCS in which students take electives from any cluster.
- **Seven genuinely new courses** enter the cluster: Generative AI, Agentic AI, MLOps, Data Engineering, Information Retrieval, Data Ethics & Security and Optimization Techniques.
- **Prerequisites disappear.** The 2025 booklet publishes no prerequisite for any specialization course. The elective pool is formally flat, and sequencing becomes entirely a departmental responsibility.
- **Topic outlines disappear too.** Only CLOs remain. This buys flexibility at the cost of comparability between universities.
- **Laboratory hours mostly disappear from the cluster:** the Data Science and AI specialization courses are listed as 3+0, where 2023 gave them three laboratory hours each.
- **Two new compulsory blocks appear:** a professional certification requirement worth three credit hours, and a three-credit field experience or internship that explicitly cannot be substituted with coursework.
- **Cloud Computing becomes compulsory** for every BS Computer Science student, in Semester VI, having had no specification at all in 2017. In the same move *System & Network Administration* and *IT Infrastructure* — compulsory for BSIT in both earlier editions — drop to optional.
- **Infrastructure gets its own specialization, and it reaches into this cluster.** *Network Infrastructure & Cloud Computing* lists twenty courses including DevOps, DevSecOps, Infrastructure as Code, Microservices and Docker, Software Defined Networks and Edge Computing — and also *Machine Learning* and *MLOps*.

i Naming drift worth tracking in your own scheme of studies

- *Advanced Statistics* (2023) → *Computational Statistics* (2025)
- *Artificial Neural Networks & Deep Learning* (2023) → *Deep Learning* (2025)
- *Data Warehousing & Business Intelligence* (2023) → *Business Intelligence + Data Engineering* (2025)
- *Tools and Techniques in Data Science* — an MS core course in 2017, an undergraduate DS elective in 2025, absent in 2023
- *Digital Image Processing* (2017, 2023) → absorbed into *Computer Vision* (2025)
- *Text Mining* and *Topics in Data Science* (2023) → dropped; nearest 2025 equivalents are *Information Retrieval* and *NLP*
- *Virtual Systems and Services* (2017, 2023) → dropped; its ground is taken by *Cloud Computing* and, in the 2025 infrastructure cluster, by *Microservices Architecture and Docker Containers*
- *IT Infrastructure* (2017) → *Information Technology Infrastructure* (2023, 2025) — same course, longer name
- *Cloud Computing* (2025 major core) and *Cloud Infrastructure and Services* (2025 IT cluster) are **two different courses** in the same booklet with heavily overlapping outcomes; see Part 5

PART 5

Where the Courses Overlap

The cluster contains real duplication. A department that offers several of these courses without deciding the boundaries will teach decision trees four times and transformers never. The table gives the overlap and the split that resolves it.

Pair	What both claim	Recommended split
Artificial Intelligence vs Machine Learning	Supervised, unsupervised and reinforcement learning are named in the AI outlines of both 2017 and 2023, and ANNs are named in the 2023 AI outline	Keep AI symbolic: search, constraint satisfaction, adversarial search, knowledge representation, logic, planning, uncertainty. Give ML everything statistical, and let AI cover learning in one week as a signpost, not a unit
Machine Learning vs Data Mining	Decision trees, naive Bayes, k -NN, SVM, k -means and hierarchical clustering appear in both 2023 outlines — roughly half of Data Mining	ML owns prediction: generalization, model selection, evaluation, ensembles, regularisation. Data Mining owns discovery: preprocessing at scale, association rules and frequent patterns, anomaly and outlier detection, graph and social network mining
Introduction to Data Science vs Machine Learning	Linear regression, k -NN, k -means and naive Bayes are in the 2023 Intro to DS outline	Intro to DS owns the process: framing, data acquisition and cleaning, EDA, feature engineering, dimensionality reduction, communication, ethics. It should demonstrate two or three models, never teach the model zoo
ANN & Deep Learning vs Deep Learning (2023 only)	Generalization theory, multi-layer perceptrons, backpropagation, CNNs and autoencoders are in both outlines	Do not offer both. Offer the 2025 merged <i>Deep Learning</i> instead
Big Data Analytics vs Data Engineering (2025)	Storage, distributed processing and pipelines	Big Data Analytics owns the distributed compute model: HDFS, MapReduce, Spark, streaming. Data Engineering owns the pipeline: ingestion, ETL, orchestration, schema and contract design, warehousing
NLP vs Text Mining vs Information Retrieval	Vector space model, text classification, precision and recall appear in all three	Offer NLP as the language course (representation, sequence models, generation); offer Information Retrieval only if search ranking and indexing are taught properly. Do not offer Text Mining as well

Pair	What both claim	Recommended split
Data Visualization vs Business Intelligence	Charts, dashboards, decision support	Visualization owns perception, chart selection and the grammar of graphics; BI owns the enterprise stack: dimensional modelling, OLAP, reporting, governance
Generative AI vs Deep Learning	GANs and VAEs are named in both 2025 CLO sets	Deep Learning owns architectures and training; Generative AI owns large pretrained models, prompting, fine-tuning, retrieval augmentation and evaluation of generated output
Cloud Computing vs Cloud Infrastructure and Services (2025)	Service models, deployment, management, compliance — these are two courses in the same booklet, one a major core and one an IT elective, with four of five outcomes in common	Do not offer both to the same student. If the IT specialization is being run, treat <i>Cloud Infrastructure and Services</i> as satisfied by the core <i>Cloud Computing</i> and spend the slot elsewhere
Cloud Computing vs Virtual Systems & Services (2017, 2023)	Hypervisors, virtual machines, resource pooling	Virtualisation is one chapter of cloud computing, not a peer of it. Where both exist in a scheme of studies, retire <i>Virtual Systems & Services</i>
System & Network Administration vs IT Infrastructure	Servers, storage, operating systems, availability, monitoring — the 2017 outlines overlap on most of the middle of both courses	Administration owns the hands-on: installation, users and groups, services, firewall rules, shell scripting, logs. Infrastructure owns the architectural: availability and performance patterns, data centres, capacity, service delivery and support processes, lifecycle
Cloud Computing vs Big Data Analytics and MLOps	Elastic compute, cluster provisioning, managed services	Cloud owns the platform and its economics: service models, provisioning, cost, security and compliance. Big Data Analytics owns the distributed compute model that runs on it; MLOps owns the model lifecycle that runs on both

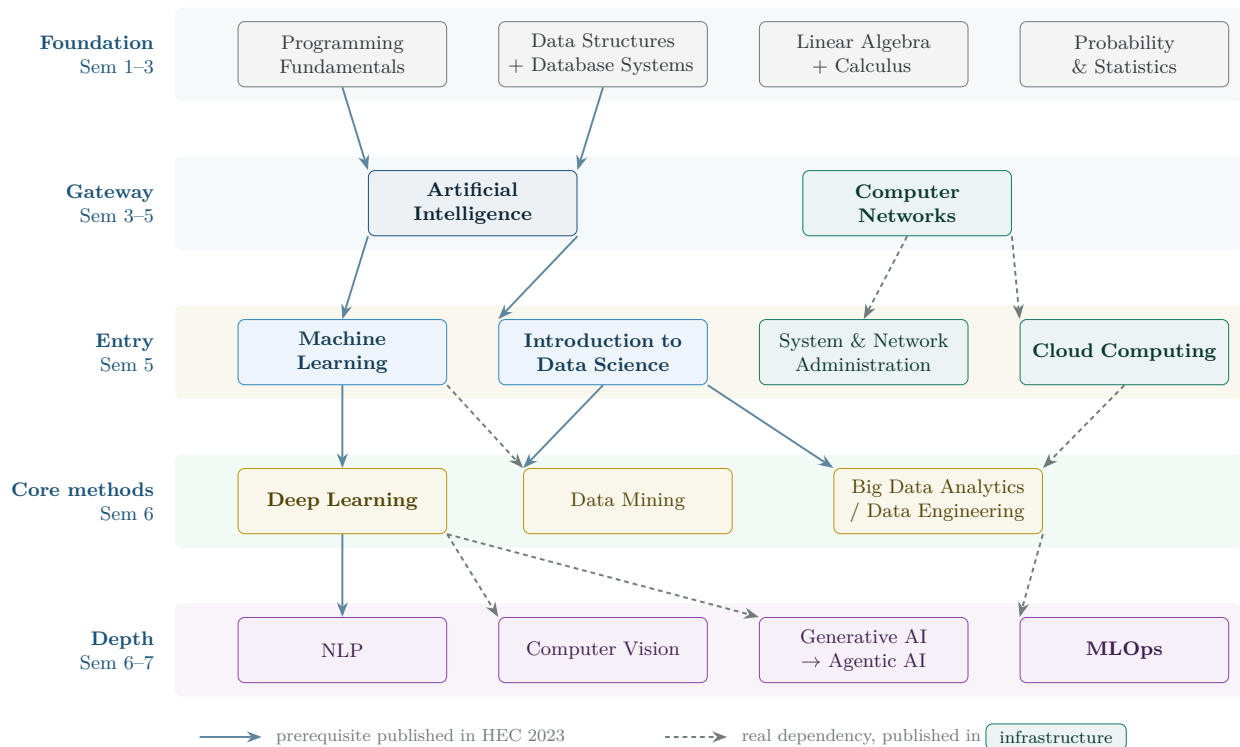
✓ The one sequencing rule that follows from this table

If a scheme of studies contains Cloud Computing *and* Big Data Analytics *and* MLOps, teach them in that order. Each one assumes the previous one's platform, and taught in the wrong order the later course spends its first three weeks re-teaching the earlier one. This is the only infrastructure sequencing constraint in the whole document, and no HEC edition states it.

PART 6

The Pathway

The diagram reconciles the three editions into one dependency graph. Solid arrows are prerequisites published in the 2023 booklet, which is the only edition that states them. Dashed arrows are dependencies that are real but unpublished — they are what a department must enforce itself, particularly under 2025, where no prerequisites are given at all.



The foundation band underpins everything below it and its arrows are omitted for legibility: linear algebra and calculus are assumed by Machine Learning and Deep Learning, and probability and statistics by every course from the Entry band down. The teal lane is the infrastructure the rest of the graph runs on: it joins the data lane at Big Data Analytics and again at MLOps, which also assumes Machine Learning. HEC 2025 publishes no prerequisite for any specialization course, so under that edition every arrow here is a departmental responsibility rather than a curricular rule. Computational Statistics, Data Visualization, Business Intelligence, Reinforcement Learning, Information Retrieval, Speech Processing and Data Ethics & Security are satellites of this graph: each hangs off one course above it and none is a prerequisite for anything else.

Figure 1: The AI, data science and infrastructure pathway, reconciled across the three HEC editions. The five bands are the order in which the material can actually be learned; the teal lane is the platform the data lane runs on. Semester ranges assume the HEC 2023 BSIT structure, in which Artificial Intelligence and Computer Networks are compulsory by Semester 3 and elective slots open in Semester 5.

6.1 Reading the pathway

- **Nothing in the cluster can start before Semester 5** for a BSIT student, because the earliest elective slots are in Semester 5 and *Artificial Intelligence* — the published prerequisite for both Machine Learning and Introduction to Data Science — is not finished until the end of Semester 3.
- **The chain has depth four:** AI → Machine Learning → Deep Learning → NLP. With elective slots only in Semesters 5, 6 and 7 this exactly fills the available room, and it is why NLP cannot sensibly be offered before the final year.

- **Two entry points, not one.** A student may enter through *Machine Learning* (the modelling route) or through *Introduction to Data Science* (the process route). The second is the better entry for IT students, because it starts from data handling and databases, which they already have.
- **The frontier tier is optional and should be treated as such.** Generative AI, Agentic AI and MLOps are genuinely valuable and genuinely hard to staff. A pathway that delivers the first three tiers well is better than one that offers all five badly.

PART 7

Networking Depth and the CCNA

The three infrastructure courses in Part 3 are the ones a BSIT student meets by default. This part asks a narrower question: if a department wants to put its graduates through Cisco's CCNA, what does the HEC curriculum already give them, and what must it add?

⚠ This part is aligned to CCNA v2.0, not v1.1

Cisco announced blueprint **version 2.0** on 20 May 2026. **Version 1.1 retires on 2 February 2027 and v2.0 goes live on 3 February 2027**, keeping the same exam number, 200-301. A student entering in Fall 2026 will not sit the exam before 2028, so *every one of them will sit v2.0*. Everything below is mapped against v2.0. Section 7.5 records what changed from v1.1 for anyone carrying an existing course file forward.

7.1 What networking courses actually exist

The first finding is that there is very little to work with before 2025.

Edition	Networking courses in the BS curriculum	What is missing
2017	<i>Computer Networks</i> (computing core, 3+1); <i>System and Network Administration</i> (BS-IT core, 3); <i>Multimedia Communications</i> (elective). <i>Wireless Network Security</i> and <i>Data Communication Networks & Security</i> exist, but only in the MS Cyber Security elective lists	No Advanced Computer Networks. No Routing & Switching. No Network Design, no Network Management, no wireless networking course. Two undergraduate networking courses in total, one of them administration
2023	<i>Computer Networks</i> (computing core, 3 (2-3)); <i>Network Security</i> ; <i>System & Network Administration</i> (BSIT domain core); <i>Wireless and Mobile Security</i>	The same gap. The master list of roughly 180 courses contains no Advanced Computer Networks and no Routing & Switching, and of the four that exist only <i>Computer Networks</i> and <i>Network Security</i> have published contents

Edition	Networking courses in the BS curriculum	What is missing
2025	<i>Computer Networks</i> and <i>Cloud Computing</i> as compulsory major courses, plus a twenty-course <i>Network Infrastructure & Cloud Computing</i> specialization: Internetworking, Network Management, Network Performance, Network Programmability and Automation, Network Automation and Orchestration, Software Defined Networks, Next-Generation Networks and Edge Computing, Enterprise Network Infrastructure Engineering Design, Cloud Design and Implementation, Wireless and Mobile Networks, DevOps, DevSecOps, MLOps, Machine Learning and more	Still no course called Routing & Switching or Advanced Computer Networks, but <i>Internetworking</i> is the functional equivalent of the first and the cluster as a whole is the equivalent of the second. For the first time there is enough material to build a networking track

💡 The honest answer to “where is Advanced Computer Networks?”

It does not exist under that name in any of the three editions, and neither does Routing & Switching. What HEC 2025 offers instead is *Internetworking* — “identify the role of routers, switches, and gateways”, “configure internetworking devices for data transmission across heterogeneous networks”, “evaluate performance trade-offs in different internetworking topologies”. That is the routing-and-switching course, and under v2.0 it is the single most important elective in the degree: domains 2.0 and 3.0 together are 45% of the exam and this is the only course that touches either. A department may of course name a locally approved course *Advanced Computer Networks* or *Routing & Switching*: HEC 2025 explicitly permits adding courses to a specialization cluster after statutory approval, and publishes no outline that a new course would contradict.

7.2 What CCNA v2.0 asks for

Five domains, twenty-nine topics, a 120-minute exam.

CCNA 200-301 v2.0 domain	Weight	What it asks for
1.0 Network Infrastructure and Connectivity	25%	Diagnosing interface and cable faults; hypervisors, virtual machines and containers; troubleshooting IPv4 and IPv6 addressing and subnetting; wireless principles; troubleshooting wired and wireless client connectivity on Windows, macOS and Linux; troubleshooting DHCPv4 client, server and relay

CCNA 200-301 v2.0 domain	Weight	What it asks for
2.0 Switching and Network Access	25%	Configuring switch-to-switch and switch-to-router connectivity — 802.1Q trunks, LACP EtherChannel, SVIs; configuring edge-host port attributes including VLAN and PoE; validating documentation with CDP and LLDP; troubleshooting with show commands, ping, traceroute and packet capture; configuring Rapid PVST+ with PortFast, root guard, loop guard and BPDU guard
3.0 IP Routing	20%	Interpreting a routing table to find the next hop; troubleshooting IPv4 and IPv6 static routing including floating statics; configuring single-area OSPFv2 and OSPFv3; interpreting HSRP and VRRP status
4.0 Network Services and Security	20%	Local usernames and AAA with TACACS+ and RADIUS; secure file transfer with SFTP and SCP; NAT and PAT on IOS XE; diagnosing DNS record faults; IPsec VPNs; configuring IPv4 access control lists; layer 2 security — DHCP snooping, dynamic ARP inspection, storm control, RA guard, port security
5.0 AI, and Network Operations and Management	10%	The role of agentic AI in network operations; selecting a prompt for a generative AI system to support network operations; network management approaches including infrastructure as code; SNMP; using Ansible to execute commands; interpreting syslog severity and facilities

💡 Read the verbs, not the weights

Counting the top-level topic statements in the published v2.0 blueprint gives a sharper picture than the percentages do. Of twenty-nine topics, **eight begin *Troubleshoot* or *Diagnose* and eight begin *Configure*** — so **55% of the exam is hands-on work at a command line**. Only six topics, 21%, are *Describe*. The remainder are *Interpret* (3), *Validate*, *Manage*, *Select* and *Use*.

For comparison, the word “troubleshoot” appears in *no* topic statement of v1.0 or v1.1. This is the change that matters, and it is not visible in the domain weights at all.

7.3 Mapping the HEC courses onto v2.0

✓ Covered 🟡 Partly covered ✗ Not covered by any published HEC outline

v2.0 domain		HEC courses that carry it	What is still missing
1.0 Infrastructure and Connectivity (25%)		<i>Computer Networks</i> in all three editions covers the addressing and subnetting knowledge behind 1.3 and 1.4, and the physical layer behind 1.1. <i>Cloud Computing</i> (2025) covers 1.2 hypervisors, virtual machines and containers outright. <i>Wireless and Mobile Networks</i> (2025) covers 1.5. <i>System and Network Administration</i> (2017 outline) covers the operating system side of 1.6	The verbs. HEC teaches addressing; v2.0 asks students to <i>troubleshoot</i> it. Nothing covers 1.7, DHCPv4 client, server and relay on IOS devices
2.0 Switching and Network Access (25%)		<i>Internetworking</i> (2025) is the only course that reaches switches at all	Everything nameable. 802.1Q trunking, LACP EtherChannel, SVIs, VLAN and PoE port configuration, CDP and LLDP, Rapid PVST+ with PortFast and the guard features, and the whole of 2.4 — show commands, extended ping, traceroute, packet capture. Not one of these terms appears in any HEC outline of any edition
3.0 IP Routing (20%)		<i>Computer Networks</i> names “routing protocols” generically; <i>Internetworking</i> (2025) covers routers and topology trade-offs	Routing-table interpretation, administrative distance, static and floating routes, single-area OSPFv2 and OSPFv3 , HSRP and VRRP. OSPFv3 is new in v2.0 and appears nowhere in HEC
4.0 Network Services and Security (20%)		<i>Network Security</i> (2023 outline) is the strongest single match in the mapping: threats, access control, firewalls, IDS/IPS, IPsec, SSL and VPN cover 4.5 and the thinking behind 4.6 and 4.7. <i>System and Network Administration</i> (2017) covers IP tables, securing network traffic, network services and logs	ACL <i>configuration</i> (4.6), AAA with TACACS+ and RADIUS (4.1), the five layer 2 security features by name (4.7), NAT and PAT (4.3), SFTP and SCP (4.2), and DNS record diagnosis (4.4)

v2.0 domain		HEC courses that carry it	What is still missing
5.0 AI, Operations and Management (10%)	✓	<i>Network Management</i> (2025) covers 5.3, 5.4 and 5.6 — management approaches, monitoring, metrics. <i>Generative AI</i> and <i>Agentic AI</i> (2025) cover 5.1 and 5.2 directly. <i>Network Programmability and Automation</i> (2025) covers 5.5 Ansible and the infrastructure-as-code half of 5.3. <i>Cloud Computing</i> covers cloud-based management	Nothing significant under 2025. Under 2017 and 2023 the entire domain is absent: no management course, no automation course, no AI in an operations context

💡 The one domain where this department has an unfair advantage

Domain 5.0 asks students to describe the role of agentic AI in network operations and to *select a prompt* for a generative AI system, with prompt components named explicitly — data classification, output format, persona, instructions. The exam description goes further and warns that candidates “may be required to evaluate output and recommendations from agentic AI and digital network assistants” anywhere in the exam, not only in domain 5.

A department already teaching the 2025 *Generative AI* and *Agentic AI* courses of Part 3 is teaching this material to a depth the exam does not approach. That is the one place where the two halves of this document meet, and it is worth saying to students explicitly: the AI courses are not a detour from the networking track, they are part of it.

7.4 The gap is practice, not knowledge

⚠️ Not one configuration topic is named in any HEC outline

Searching every published HEC course outline in all three editions for the vocabulary of the exam returns almost nothing. VLAN, trunk, spanning tree, EtherChannel, OSPF, static route, NAT, DHCP, SNMP, syslog, ACL, SSH, CLI — none of them appears. Of the whole CCNA vocabulary only CIDR and IPv6 are named, both in *Computer Networks*. No HEC networking outline names a vendor, a command line, or a device configuration task.

Under v1.1 this was survivable, because much of that exam could be answered from protocol knowledge. Under v2.0 it is not: **55% of the topics are configure-or-troubleshoot**, and a student who has never opened a command line cannot answer them from theory. HEC networking courses are protocol-and-concept courses; v2.0 is a configuration-and-diagnosis exam. Closing the gap is a laboratory decision, not a syllabus one.

✓ What to actually do

1. **Put the laboratory back.** HEC 2025 lists the infrastructure courses as (2+1)/3, meaning the laboratory hour is optional. For a CCNA track it is not optional. Offer every networking course as 2+1, never 3+0.
2. **Write the verbs into the local CLOs.** Because HEC 2025 publishes no outlines, a department is free — and now obliged — to write “configure” and “diagnose a broken configuration” into its own course learning outcomes rather than “explain” and “describe”. This is the single highest-leverage change available, and it costs nothing.
3. **Standardise on one simulator across the strand.** Packet Tracer for the first two courses, then GNS3 or CML where IOS XE behaviour matters — 4.3 names IOS XE explicitly. Consistency matters more than which tool.

4. **Teach show commands as a subject, not an aside.** Topic 2.4 makes reading **show** output, extended ping, traceroute and packet capture examinable in its own right. It is the skill that underpins all eight troubleshooting topics.
5. **Spend the certification requirement on CCNA itself.** HEC 2025 makes three credit hours of professional certification a condition of degree completion, and CCNA is worth far more than the sixteen study hours per credit the booklet asks for.
6. **Use the field experience as a network operations placement.** The three-credit internship cannot be substituted with coursework, so a NOC, ISP or campus network team turns a requirement into exam preparation.

✓ The scheme that follows from all this

Scheme E in Part 8 turns this mapping into an eight-elective track: six networking courses, two AI courses because domain 5.0 now examines agentic AI and prompt selection, and the compulsory professional-certification credit spent on CCNA itself. It sits with the other four schemes so that the choice between a data track and a networking track can be made side by side.

7.5 If you are carrying a v1.1 course file forward

The changes from v1.1 to v2.0 are large enough that an existing alignment document should be rewritten rather than patched.

Change	What it means for a scheme of studies
Six domains become five	Network Fundamentals and Network Access are re-cut into Network Infrastructure and Connectivity (25%) and Switching and Network Access (25%). Core networking and switching are now half the exam
The automation domain disappears	Folded into the new 5.0 alongside AI and management. REST APIs, JSON and Terraform are gone; Ansible survives as 5.5. A dedicated automation elective is no longer essential
Troubleshooting arrives	From zero topic statements to eight. With eight more beginning <i>Configure</i> , 55% of the exam is hands-on. This is the change that should drive laboratory provision
A real AI section appears	Agentic AI and prompt engineering are examinable, and agentic-AI output may appear anywhere in the exam. Favours a department that teaches the Part 3 AI courses
Added	OSPFv3 for IPv6; hypervisors and containers; SFTP and SCP; DNS record diagnosis; storm control and RA guard; PoE port configuration; DHCP server and relay configuration; SVIs
Removed	QoS; NTP; TFTP and FTP; the OSI and TCP/IP model descriptions; TCP versus UDP; device-role and topology-architecture theory; switching concepts such as MAC learning; WLC GUI configuration; security programme elements and password policy; REST API architecture; JSON; Terraform

Doing this under HEC 2023 instead

It is possible but much thinner. BSIT 2023 gives *Computer Networks* in Semester 3 and *System & Network Administration* as a domain core, then seven domain electives of which only *Network Security* is a named networking option. Nothing in the 2023 curriculum touches domains 2.0, 3.0 or 5.0 — 55% of the v2.0 exam. A department would have to add locally approved courses, and since HEC 2023 publishes no contents for the IT cluster anyway, writing those courses locally is no more of a departure than teaching the ones already listed. The realistic 2023 track is: Computer Networks (core), System & Network Administration (core), Network Security (elective), plus three locally approved courses named Routing & Switching, Network Operations and Advanced Computer Networks. That is a curriculum change requiring statutory approval, which is why Scheme E above is written against 2025.

PART 8

Semester-Wise Schemes

Five schemes follow. The first four build a data science track; the fifth, carried over from Part 7, builds a networking track aimed at CCNA. They are a menu rather than a sequence — a department picks one.

All of them leave the compulsory parts of the degree untouched and spend only elective credit, so none requires HEC approval of a new programme. They are choices a department makes within a curriculum it already has.

8.1 Scheme A — BSIT under HEC 2023, full data science track

Seven domain electives, twenty-one credit hours, all spent on the cluster. This produces a genuine concentration and satisfies every prerequisite the 2023 booklet publishes. It also spends the entire elective budget, which is the trade-off discussed after the table.

Sem	Course	Credits	Status and prerequisite
1–2	Programming Fundamentals; Object Oriented Programming; Database Systems; Multivariable Calculus; Linear Algebra	as per BSIT	Already compulsory. No change
3	Data Structures; Artificial Intelligence ; Probability & Statistics	as per BSIT	Already compulsory. <i>Artificial Intelligence</i> is the gateway, and it is already in the core
4	—	—	No elective slots. Domain cores only
5	Introduction to Data Science (Domain Elective 1)	3 (2-3)	Prerequisite: Artificial Intelligence ✓ Sem 3
5	Machine Learning (Domain Elective 2)	3 (2-3)	Prerequisite: Artificial Intelligence ✓ Sem 3
6	Data Mining (Domain Elective 3)	3 (2-3)	Prerequisite: Introduction to Data Science ✓ Sem 5
6	Deep Learning (Domain Elective 4)	3 (2-3)	Prerequisite: Machine Learning ✓ Sem 5

Sem	Course	Credits	Status and prerequisite
6	Big Data Analytics (Domain Elective 5)	3 (2-3)	Prerequisite: Introduction to Data Science ✓ Sem 5
6	Data Visualization (Domain Elective 6)	3 (2-3)	Pairs with Data Mining; no hard prerequisite
7	Natural Language Processing (Domain Elective 7)	3 (2-3)	Prerequisite: Deep Learning ✓ Sem 6. Substitute Generative AI or MLOps if staffing favours it
7–8	Final Year Project I and II	2 + 4	Already compulsory. Supervise within the strand

⚠ What Scheme A costs

Spending all seven domain electives on the cluster means a BSIT graduate takes *no* Cloud Computing, Network Security, Web Engineering, Enterprise Systems or Virtual Systems & Services — the courses the 2023 booklet itself names as BSIT elective examples. **Losing Cloud Computing is the one that hurts**, because Big Data Analytics in Semester 6 then has to teach its own platform from scratch, and because HEC 2025 has since made Cloud Computing compulsory for every computing student. Computer Networks (Semester 3), System & Network Administration and IT Infrastructure survive — they are BSIT domain *core*, not electives — so the graduate is not left without infrastructure entirely. That may be the right call for a department building a data specialization, but it should be a decision, not an accident. Scheme B below is the balanced alternative.

8.2 Scheme B — BSIT under HEC 2023, balanced track (recommended)

Four of the seven electives on the cluster, three left for IT depth. This is the scheme to adopt if the department wants every BSIT graduate to be competent with data without turning the degree into a data science degree.

Sem	Course	Credits	Note
3	Artificial Intelligence (core); Computer Networks (core)	3 (2-3) each	Both compulsory already. The two gateways, one per lane
5	Introduction to Data Science (DE 1)	3 (2-3)	The entry point. Process, EDA, feature work, ethics
5	Cloud Computing (DE 2)	3 (2-3)	The platform course. Put it here, not later: Big Data Analytics and MLOps both assume it, and HEC 2025 has since made it compulsory for every computing student
5	<i>(System & Network Administration is a domain core in Semester 5 — it costs no elective credit)</i>	3 (2-3)	Follows Computer Networks. Already compulsory for BSIT
6	Machine Learning (DE 3)	3 (2-3)	Prerequisite: Artificial Intelligence ✓ Sem 3

Sem	Course	Credits	Note
6	Big Data Analytics (DE 4)	3 (2-3)	Prerequisite: Introduction to Data Science ✓ Sem 5. Now has its platform from Cloud Computing ✓ Sem 5
6	IT elective — e.g. Network Security (DE 5)	3 (2-3)	Keeps the IT identity of the degree
6	IT elective — e.g. Web Engineering (DE 6)	3 (2-3)	
7	Deep Learning (DE 7)	3 (2-3)	Prerequisite: Machine Learning ✓ Sem 6. Substitute Data Mining if the departmental strength is statistics rather than neural networks
7–8	Final Year Project I and II	2 + 4	At least one project topic per supervisor drawn from the strand

✓ Why this is the default recommendation

Four cluster courses — Introduction to Data Science, Machine Learning, Big Data Analytics and Deep Learning — is the smallest set that produces a graduate who can frame a data problem, build and evaluate a model, run it at scale and understand what a neural network is doing. The fifth slot goes to **Cloud Computing**, which is not a concession to the IT side of the degree but a prerequisite for the data side: it is what Big Data Analytics runs on, and it is the course HEC 2025 later made compulsory for everyone. Two IT electives still survive on top of a domain core that already contains Computer Networks, System & Network Administration and IT Infrastructure, so the degree still reads clearly as Information Technology.

8.3 Scheme C — BSCS with an IT specialization under HEC 2025

This is the harder case. The 2025 booklet's *Information Technology* specialization cluster contains fifteen courses — Web Technologies, Cyber Security, Database Administration, System & Network Administration, IT Infrastructure, Network Security, Data Communication, Wireless & Mobile Networks, Cloud Infrastructure, Email Systems, Ethical Hacking, Security Policies, IT Project Management, Systems Analysis & Design and Technology Lifecycle Planning — and *not one of them is an AI or data course*. Taking the named IT specialization strictly means taking none of this cluster.

💡 Three ways out, and the booklet supplies all of them

First, universities may run a **generic BS Computer Science** in which students take electives from any cluster, with no specialization named on the degree or transcript. Second, HEIs **may add courses to a cluster** after approval of their own statutory bodies — this route keeps the specialization title. Third, and most simply, **specialization 12, Network Infrastructure & Cloud Computing, already contains Machine Learning and MLOps** alongside DevOps, DevSecOps, Infrastructure as Code, Software Defined Networks, Microservices and Docker, and Edge Computing. For a student who wants infrastructure with real data content, that named specialization is a better fit than the Information Technology one and needs no local approval at all.

i Two things HEC 2025 gives this student for free

Computer Networks is a major core course in Semester V and **Cloud Computing** is a major core course in Semester VI. Neither costs an elective slot. That is a real improvement on 2023 for anyone building a data track: the platform half of the pathway arrives compulsorily, and all eight elective slots stay available. It also means *Cloud Infrastructure and Services* in the IT cluster is largely redundant for these students — see Part 5.

Sem	Course	Credits	Note
I-IV	Programming Fundamentals; OOP; Digital Logic Design; Data Structures; Database Systems; Operating Systems; Software Engineering; Computer Organization; Design & Analysis of Algorithms	as per BSCS	Compulsory. Calculus is IDS-I, Linear Algebra is IDS-II
V	Artificial Intelligence; Computer Networks (Major core)	3 each	Both compulsory, and both free of elective credit. The two gateways, but two years later than under 2023
V	Probability & Statistics as IDS-III	3	Must be chosen deliberately — see the warning below
VI	Cloud Computing (Major core)	3	Compulsory. Costs no elective slot, and arrives exactly when the data electives start needing a platform
VI	Introduction to Data Science (Elective I)	3	DS cluster
VI	Machine Learning (Elective II)	3	DS, AI <i>and</i> Network Infrastructure clusters
VI	System & Network Administration (Elective III)	2+1	IT cluster
VI	Web Technologies (Elective IV)	2+1	IT cluster
VII	Deep Learning (Elective V)	3	DS and AI clusters
VII	Big Data Analytics (Elective VI)	3	Its platform is already in place from Semester VI
VII	MLOps (Elective VII)	3	Where the two lanes meet. In the AI <i>and</i> Network Infrastructure clusters, so it counts either way
VII	IT Infrastructure (Elective VIII)	2+1	IT cluster. Do not also take <i>Cloud Infrastructure and Services</i> — the core Cloud Computing course already covers it

Sem	Course	Credits	Note
VII	Professional Certification	3	Compulsory in 2025. Align it with the strand — the booklet’s own AI specialization annexe recommends MOOC routes for exactly these courses
VIII	Final Year Project + Field Experience	6 + 3	Both compulsory. The internship cannot be substituted with coursework

The probability trap in HEC 2025

Under HEC 2023, Probability & Statistics is a required mathematics course in Semester 3 for both BSIT and BSCS. Under HEC 2025 it is **not compulsory**: only Calculus and Linear Algebra are mandatory interdisciplinary courses. Probability & Statistics is item four in an optional pool of eighteen, and the two free interdisciplinary slots (IDS-III and IDS-IV) both fall in **Semester V**. A student who does not choose it will arrive at Machine Learning in Semester VI with no statistics at all. Any department running a data science or AI specialization under the 2025 curriculum should make Probability & Statistics the mandated IDS-III for those students, and should consider petitioning to move it earlier.

8.4 Scheme D — the minimum viable strand

For a department that cannot yet staff a full track, three courses in this order still produce a usable graduate. Everything else can wait.

#	Course	What the graduate can do that they could not before
1	Introduction to Data Science	Frame a question as a data problem, audit and clean a dataset, explore it honestly, and know when the answer is not in the data
2	Machine Learning	Build a predictive model, choose the right metric, avoid leakage, and read a result critically
3	One applied course — Big Data Analytics, Data Mining or Deep Learning	Work at the scale, or with the model family, that the local job market actually asks for

8.5 Scheme E — a CCNA v2.0 track under HEC 2025

Six of the eight electives on networking, two on the AI side because domain 5.0 now examines it. Computer Networks and Cloud Computing are compulsory major courses and cost no elective credit at all.

Sem	Course	Credits	v2.0 topics served
V	Computer Networks (major core)	3	The knowledge behind 1.1, 1.3 and 1.4. Add subnetting drill and interface diagnosis to the laboratory

Sem	Course	Credits	v2.0 topics served
V	Probability & Statistics as IDS-III	3	Not CCNA. Required for the data electives — see the warning in Part 8
VI	Cloud Computing (major core)	3	1.2 hypervisors, virtual machines and containers; the cloud-based half of 5.3
VI	Internetworking (Elective I)	2+1	2.1, 2.2, 2.5, 3.1, 3.2, 3.3, 3.4. The load-bearing course of the whole track. Write VLANs, 802.1Q, EtherChannel, SVIs, Rapid PVST+, static routing and OSPFv2/v3 into its local CLOs
VI	System & Network Administration (Elective II)	2+1	1.6 client connectivity on Windows, macOS and Linux; 1.7 DHCP; 4.1 local users and AAA; 4.2 secure file transfer
VI	Machine Learning (Elective III)	3	Underpins domain 5.0 and the wider AI agenda. Not directly examinable
VII	Network Security (Elective IV)	2+1	4.5, 4.6, 4.7. Add ACL configuration and the five layer 2 security features by name — DHCP snooping, DAI, storm control, RA guard, port security
VII	Network Management (Elective V)	2+1	5.3, 5.4, 5.6 — management approaches, SNMP, syslog severity and facilities
VII	Generative AI (Elective VI)	3	5.1, 5.2 — agentic AI in operations, prompt selection and components
VII	Wireless and Mobile Networks (Elective VII)	2+1	1.5 wireless principles; the wireless half of 1.6; 2.2.b access points
VII	Enterprise Network Infrastructure Engineering Design (Elective VIII)	2+1	The design and troubleshooting capstone. Run 2.4 — show commands, extended ping, traceroute, packet capture — as its spine
VII	Professional Certification: CCNA	3	Compulsory in HEC 2025. This is where the exam itself sits
VIII	Final Year Project + Field Experience	6 + 3	Place the internship in a NOC, ISP or campus network team

i Two substitutions worth considering

Swap *Wireless and Mobile Networks* for *Network Programmability and Automation* if the department has automation strength: it picks up 5.5 Ansible and the infrastructure-as-code

half of 5.3, at the cost of 1.5 and 2.2.b. Swap *Machine Learning for Network Performance* if the cohort has no interest in the data side at all — but note that this leaves domain 5.0 resting on *Generative AI* alone, and that Machine Learning appears in the *Network Infrastructure & Cloud Computing* cluster itself, so it counts towards the specialization either way.

PART 9

Risks and Implementation Notes

9.1 Things that will go wrong if they are not planned

1. **Prerequisites will not enforce themselves under HEC 2025.** The booklet publishes none. Adopt the 2023 chains explicitly in your own scheme of studies and encode them in the student information system.
2. **Laboratory hours will quietly disappear.** 2023 gave every cluster course three laboratory hours; 2025 lists them as 3+0. This cluster cannot be taught from a whiteboard. If you adopt the 2025 structure, write the laboratory back in as 2+1.
3. **Machine Learning and Data Mining will be taught twice.** Assign the boundary in Part 5 to named course owners before the first offering, not after the first student complaint.
4. **The NLP course will be a decade out of date.** No edition's published NLP specification mentions embeddings, attention, transformers or language models. The course code and title are fine; the content must be written locally.
5. **Statistics will be missing.** See the warning above for HEC 2025. Even under 2023, one Probability & Statistics course in Semester 3 is thin support for Semester 6 machine learning; a Computational Statistics elective is worth the slot if one is free.
6. **The frontier courses will be assigned to whoever is available.** Generative AI, Agentic AI and MLOps have no published outline in any edition. Do not offer one without a named owner who has built something in it.
7. **The IT courses have no syllabus to inherit.** HEC 2023 specifies the AI, Data Science and Cyber Security clusters and leaves the Information Technology one completely unspecified; HEC 2025 gives every specialization course five outcome statements and nothing else. For System & Network Administration, IT Infrastructure and Virtual Systems & Services, the 2017 outlines are the only detailed descriptions HEC has ever published, and they are still broadly sound. Use them rather than starting from a blank page.
8. **Cloud Computing will be scheduled too late.** It is the platform for Big Data Analytics, Data Engineering and MLOps, so a scheme that puts it in the final year forces each of those courses to teach its own infrastructure from scratch. Put it no later than the semester before the first of them.
9. **Cloud will be taught twice under HEC 2025.** *Cloud Computing* is a compulsory major course and *Cloud Infrastructure and Services* is an IT specialization elective, and four of their five outcomes are the same. Bar the combination in the scheme of studies.

9.2 What to do with the 2025 certification requirement

The 2025 booklet makes three credit hours of professional certification a condition of degree completion, and its own annexe recommends MOOC routes aligned to the AI specialization — a foundational block (probability, discrete mathematics, data structures) and a core AI and machine learning block mapped course by course to Coursera and edX offerings. For a student on this pathway the certification block is best spent reinforcing the strand rather than diversifying away from it: a

machine learning or cloud data engineering certification completed alongside Semester VII electives both satisfies the requirement and strengthens the final year project.

9.3 A note on the three editions coexisting

Universities do not switch curricula cleanly. In practice a department may be running 2023 for its current cohorts, planning 2025 for the next intake, and still carrying 2017-era course files in its quality-assurance records. When that happens the mapping that matters is the one in Part 2: it lets a course approved under one edition be recognised under another. Two rules make the mapping safe. Match on *outline*, not on title — *Advanced Statistics* and *Computational Statistics* are the same course, while the 2017 and 2025 courses both called *Natural Language Processing* are not. And where an edition splits a course, treat the pair as one — a student who took 2023's *ANN & Deep Learning* has met 2025's *Deep Learning* requirement.

PART 10

The Published Course Contents

Everything below is what the three booklets actually say, reproduced rather than summarised, so that a department writing a course file can lift the wording directly and see at a glance which edition gives it most to work with.

Three things to know before reading. **HEC 2017** publishes credit hours, a prerequisite, CLOs with Bloom levels, a topic outline and reference books for all seventy-five bachelor courses it lists. **HEC 2023** publishes the same, plus a course-introduction paragraph, but *only for forty-six courses* — those common to every computing degree, plus the Artificial Intelligence, Data Science and Cyber Security cluster tables. **HEC 2025** publishes course learning outcomes and nothing else: no prerequisite, no outline, no reading list, for any of its two hundred and fifteen specialization courses.

Where an edition says nothing, this is recorded rather than passed over — the gaps are as informative as the text, and several of them are the reason a department has to write the course itself.

10.1 The AI and data science courses

Artificial Intelligence — the gateway course

2017 3+1 • prerequisite: Data Structures and Algorithms

Introduction (Introduction, basic component of AI, Identifying AI systems, branches of AI, etc.); Reasoning and Knowledge Representation (Introduction to Reasoning and Knowledge Representation, Propositional Logic, First order Logic); Problem Solving by Searching (Informed searching, Uninformed searching, Local searching.); Constraint Satisfaction Problems; Adversarial Search (Min-max algorithm, Alpha beta pruning, Game-playing); Learning (Unsupervised learning, Supervised learning, Reinforcement learning); Uncertainty handling (Uncertainty in AI, Fuzzy logic); Recent trends in AI and applications of AI algorithms (trends, Case study of AI systems, Analysis of AI systems)

Set texts: Stuart Russell and Peter Norvig, *Artificial Intelligence. A Modern Approach*, 3rd edition, Prentice Hall, Inc., 2010; Hart, P.E., Stork, D.G. and Duda, R.O., 2001. *Pattern classification*. John Willey & Sons; Luger, G.F. and Stubblefield, W.A., 2009. *AI algorithms, data structures, and idioms in Prolog, Lisp, and Java*. Pearson Addison-Wesley.

2023 3 (2-3) • prerequisite: Object Oriented Programming

An Introduction to Artificial Intelligence and its applications towards Knowledge Based Systems; Introduction to Reasoning and Knowledge Representation, Problem Solving by Searching (Informed searching, Uninformed searching, Heuristics, Local searching, Min- max algorithm, Alpha beta pruning, Game-playing); Case Studies: General Problem Solver, Eliza, Student, Macsyma; Learning from examples; ANN and Natural Language Processing; Recent trends in AI and applications of AI algorithms. Python programming language will be used to explore and illustrate various issues and techniques in Artificial Intelligence.

2025 3 • no prerequisite and no outline published

- Describe core AI concepts including search algorithms, knowledge representation, and reasoning.
- Implement basic AI algorithms for problem-solving and decision-making.
- Analyze the applications of AI in real-world scenarios.
- Discuss ethical considerations and limitations of AI systems.
- Develop simple AI models using appropriate tools and frameworks.

Programming for Artificial Intelligence — the AI toolchain course**2017 not specified**

The course does not exist in HEC 2017 at either level.

2023 3 (2-3) • prerequisite: Artificial Intelligence

The first objective of the course is to introduce and then build the proficiency of students in different AI programming languages. The basics include IDE for the languages, variables, expressions, operands and operators, loops, control structures, debugging, error messages, functions, strings, lists, object-oriented constructs and basic graphics in the languages. Special emphasis is given to writing production quality clean code in the programming language. Once the classical programming languages are properly introduced, the course should introduce some libraries necessary for interpreting, analyzing and plotting numerical data in Python (e.g., NumPy, Matplotlib, Anaconda and Pandas for Python) and give examples of each library using simple use cases and small case studies.

Set texts: Russell, S. and Norvig, P. "Artificial Intelligence. A Modern Approach", 3rd ed, Prentice Hall, Inc., 2015; Norvig, P., "Paradigms of Artificial Intelligence Programming: Case studies in Common Lisp", Morgan Kaufman Publishers, Inc., 1992; Luger, G.F. and Stubblefield, W.A., "AI algorithms, data structures, and idioms in Prolog, Lisp, and Java", Pearson Addison-Wesley. 2009.

2025 3+0 • no prerequisite and no outline published

- Write efficient and well-documented code.
- Utilize version control (Git) to manage and collaborate on AI-related code repositories.
- Implement basic Machine Learning models using standard ML frameworks

Machine Learning — the pivot of the whole cluster**2017 not specified**

No undergraduate course of this name exists in HEC 2017. It is a core course of the MS Data Science programme, which publishes no outline for it.

2023 3 (2-3) • prerequisite: Artificial Intelligence

Introduction to machine learning; concept learning: General-to-specific ordering of hypotheses, Version spaces Algorithm, Candidate elimination algorithm; Supervised Learning: decision trees, Naive Bayes, Artificial Neural Networks, Support Vector Machines, Overfitting, noisy data, and pruning, Measuring Classifier Accuracy; Linear and Logistic regression; Unsupervised Learning: Hierarchical Agglomerative Clustering, k-means partitional clustering; Self-Organizing Maps (SOM) k-Nearest-neighbor algorithm; Semi-supervised learning with EM using labeled and unlabelled data; Reinforcement Learning: Hidden Markov models, Monte Carlo inference Exploration vs. Exploitation Trade-off, Markov Decision Processes; Ensemble Learning: Using committees of multiple hypotheses. Bagging, boosting.

Set texts: Machine Learning, Tom, M., McGraw Hill, 1997; Machine Learning: A Probabilistic Perspective, Kevin P. Murphy, MIT Press, 2012.

2025 3+0 (2+1 in the SE cluster) • no prerequisite and no outline published

- Explain core concepts of Machine Learning. (C2: Understand)
- Apply machine learning algorithms to real-world datasets. (C3: Apply)
- Analyze the performance of ML models using evaluation metrics. (C4: Analyze)
- Implement ML solutions using modern toolkits. (C3: Apply)
- Discuss ethical considerations while applying ML to societal problems. (A3: Valuing)

Artificial Neural Networks & Deep Learning — the 2023 first half of deep learning**2017 not specified**

The course does not exist in HEC 2017 at either level.

2023 3 (2-3) • prerequisite: Programming for Artificial Intelligence

Introduction and history of neural networks, Basic architecture of neural networks, Perceptron and Adaline (Minimum Error Learning) for classification. Basics of deep learning, learning networks, Shallow vs. Deep learning etc.; Machine learning theory – training and test sets, evaluation, etc. Selected topics from: Gradient descent (Delta) rule, Hebbian, Neo-Hebbian and Differential Hebbian Learning, Drive Reinforcement Theory, Kohonen Self Organizing Maps, Associative memory, Bi-directional associative memory (BAM), Energy surfaces, The Boltzmann machines, Backpropagation Networks, Feedforward Networks; Theory of Generalization; Multi-layer perceptrons, error back-propagation; Deep convolutional networks, Computational complexity of feed forward and deep convolutional neural networks; Unsupervised deep learning including auto-encoders; Deep belief networks; Restricted Boltzmann Machines; Deep Recurrent Neural Networks (BPTT, LSTM, etc.); GPU programming for deep learning CuDNN; Generative adversarial networks (GANs); Sparse coding and auto-encoders; Data augmentation, elastic distortions, data normalization; Mitigating overfitting with dropout, batch normalization, dropconnect; Novel architectures, ResNet, GoogleNet, etc

Set texts: Neural Network Design, 2nd Edition, Martin T. Hagan, Howard, B. Demuth, Mark Hudson Beale and Orlando De Jesus, Publisher: Martin Hagan; 2 edition (September 1, 2014), ISBN-10: 0971732116; An Introduction to Neural Networks, James A Anderson, Publisher: A Bradford Book (March 16, 1995), ISBN-10: 0262011441; Fundamentals of Artificial Neural Networks, Mohammad Hassoun, Publisher: A Bradford Book (January 1, 2003), ISBN-10: 0262514672.

2025 3+0 • no prerequisite and no outline published

- Explain structure, function, and types of deep neural networks.
- Apply deep learning techniques such as CNNs, RNNs, and transfer learning for AI applications.
- Analyze and optimize neural networks for performance using forward and backward propagation.
- Demonstrate applications of deep reinforcement learning using modern tools.

Deep Learning — the 2023 second half, and the whole of it in 2025

2017 not specified

MS Data Science specialization core only; no outline published.

2023 3 (2-3) • prerequisite: Artificial Neural Networks and Deep Learning

Basics of deep learning, learning networks, Shallow vs. Deep learning etc.; Machine learning theory – training and test sets, evaluation, etc. Theory of Generalization; Multi-layer perceptrons, error back-propagation; Deep convolutional networks, Computational complexity of feed forward and deep convolutional neural networks; Unsupervised deep learning including auto-encoders; Deep belief networks; Restricted Boltzmann Machines; Deep Recurrent Neural Networks (BPTT, LSTM, etc.); GPU programming for deep learning CuDNN; Generative adversarial networks (GANs); Sparse coding and auto-encoders; Data augmentation, elastic distortions, data normalization; Mitigating overfitting with dropout, batch normalization, dropconnect; Novel architectures, ResNet, GoogleNet, etc

Set texts: Deep Learning by Ian Goodfellow, Yoshua Bengio, Aaron Courville (; Deep learning with python by Françoise Chollet, ISBN-10: 9781617294433, 2017.

2025 3+0 • no prerequisite and no outline published

- Understand the fundamentals of neural networks in AI C2 (Understand)
- Explain how simple ANNs can be designed C2 (Understand)
- Apply deep learning algorithms to real-world classification problems C3 (Apply)
- Analyze results from deep learning to select appropriate solutions C4 (Analyze)

Introduction to Data Science — the process-and-EDA course

2017 not specified

The nearest equivalent, Tools and Techniques in Data Science, is an MS core course with no published outline.

2023 3 (2-3) • prerequisite: Artificial Intelligence

Introduction: What is Data Science? Big Data and Data Science hype, Datafication, Current landscape of perspectives, Skill sets needed; Statistical Inference: Populations and samples, Statistical modeling, probability distributions, fitting a model, Intro to Python; Exploratory Data Analysis and the Data Science Process; Basic Machine Learning Algorithms: Linear Regression, k-Nearest Neighbors (k-NN), k-means, Naive Bayes; Feature Generation and Feature Selection; Dimensionality Reduction: Singular Value Decomposition, Principal Component Analysis; Mining Social-Network Graphs: Social networks as graphs, Clustering of graphs, Direct discovery of communities in graphs, Partitioning of graphs, Neighborhood properties in graphs; Data Visualization: Basic principles, ideas and tools for data visualization; Data Science and Ethical Issues: Discussions on privacy, security, ethics, Next-generation data scientists.

Set texts: Foundations of data science, Blum, A., Hopcroft, J., & Kannan, R., Vorabversion eines Lehrbuchs, 2016; An Introduction to Data Science, Jeffrey S. Saltz, Jeffrey M. Stanton, SAGE Publications, 2017; Python for everybody: Exploring data using Python 3, Severance, C.R., CreateSpace Independent Pub Platform. 2016.

2025 3+0 • no prerequisite and no outline published

- Understand the fundamental principles of data science. C2 (Understand)
- Apply EDA and the Data Science processes. C3 (Apply)
- Apply basic machine learning algorithms to solve real-world problems of moderate complexity. C3 (Apply)

Data Mining — pattern discovery

2017 not specified

The course does not exist at any level in HEC 2017.

2023 3 (2-3) • prerequisite: Advance Statistics, Introduction to Data Science

Introduction to data mining and basic concepts, Pre-Processing Techniques & Summary Statistics, Association Rule mining using Apriori Algorithm and Frequent Pattern Trees, Introduction to Classification Types, Supervised Classification (Decision trees, Naïve Bayes Classification, K-Nearest Neighbors, Support Vector Machines etc.), Unsupervised Classification (K Means, K Median, Hierarchical and Divisive Clustering, Kohonen Self Organizing maps), outlier & anomaly detection, Web and Social Network Mining, Data Mining Trends and Research Frontiers. Implementing concepts using Python

Set texts: Jiawei Han & Micheline Kamber, Jian Pei (2011). Data Mining: Concepts and Techniques, 3rd Edition; Pang-Ning Tan, Michael Steinbach, and Vipin Kumar (2005). Introduction to Data Mining; Charu C. Aggarwal (2015). Data Mining: The Textbook.

2025 3+0 • no prerequisite and no outline published

- Explain fundamentals of data mining concepts, techniques, and algorithms.C2 (Understand)
- Apply data mining algorithms to real-world datasets using appropriate tools and techniques.C3 (Apply)
- Evaluate and interpret the performance of data mining models for informed decision-making.C5 (Evaluate) 3. ARTIFICIAL INTELLIGENCE ELECTIVES:

Big Data Analytics — the distributed compute course

2017 3(2,1) • prerequisite: Probability and Statistics, Programming Fundamentals

Introduction to Big Data Analytics, Big Data Platforms, Data Store & Processing using Hadoop, Big Data Storage and Analytics, Big Data Analytics ML Algorithms, Recommendation, Clustering, and Classification, Linked Big Data: Graph Computing and Graph Analytics, Graphical Models and Bayesian Networks, Big Data Visualization, Cognitive Mobile Analytics.

Set texts: Mining of Massive Datasets, Jure Leskovec, Anand Rajaraman, Jeff Ullman, 2nd edition, 2011; Hadoop: The Definitive Guide, Tom White, 4th edition. 2009; Data-Intensive Text Processing with Map Reduce, Jimmy Lin and Chris, 2010.

2023 3 (2-3) • prerequisite: Introduction to Data Science

Introduction and Overview of Big Data Systems; Platforms for Big Data, Hadoop as a Platform, Hadoop Distributed File Systems (HDFS), MapReduce Framework, Resource Management in the cluster (YARN), Apache Scala Basic, Apache Scala Advances, Resilient Distributed Datasets (RDD), Apache Spark, Apache Spark SQL, Data analytics on Hadoop / Spark, Machine learning on Hadoop / Spark, Spark Streaming, Other Components of Hadoop Ecosystem

Set texts: White, Tom. "Hadoop: The definitive guide." O'Reilly Media, Inc., 2012; Karau, Holden, Andy Konwinski, Patrick Wendell, and Matei Zaharia. "Learning spark: lightning-fast big data analysis." O'Reilly Media, Inc., 2015; Miner, Donald, and Adam Shook. "MapReduce design patterns: building effective algorithms and analytics for Hadoop and other systems." O'Reilly Media, Inc., 2012..

2025 3+0 • no prerequisite and no outline published

- Understand the fundamental concepts of Big Data and its programming paradigm. C3(Understand)
- Hadoop/MapReduce Programming, Framework, and Ecosystem C2 (Understand)
- Apache Spark Programming C3 (Apply)

Advanced / Computational Statistics — the statistical support course

2017 not specified

The course does not exist in HEC 2017 at either level.

2023 3 (3-0) • prerequisite: Probability and Statistics

Introduction to Statistics, Use of Statistics in Data Science, Experimental Design, Statistical Techniques for Forecasting, Interpolation/ Extrapolation, Introduction to Probability, Conditional Probability, Prior and Posterior Probability, Random number generation (RNG), Techniques for RNG, Correlation analysis, Chi Square Dependency tests, Diversity Index, Data Distributions Multivariate Distributions, Error estimation, Confidence Intervals, Linear transformations, Gradient Descent and Coordinate Descent, Likelihood inference, Revision of linear regression and likelihood inference, Fitting algorithms for nonlinear models and related diagnostics, Generalized linear model; exponential families; variance and link functions, Proportion and binary responses; logistic regression, Count data and Poisson responses; log-linear models, Overdispersion and quasi-likelihood; estimating functions, Mixed models, random effects, generalized additive models and penalized regression; Introduction to SPSS, Probability/ Correlation analysis/ Dependency tests/ Regression in SPSS.

Set texts: *Probability and Statistics for Computer Scientists*, 2nd Edition, Michael Baron; *Probability for Computer Scientists*, online Edition, David Forsyth; *Discovering Statistics using SPSS for Windows*, Andy Field.

2025 3+0 • no prerequisite and no outline published

- Understand the fundamental statistics for data science and applications of statistics in data science.C1 (Knowledge)
- Apply Statistical techniques in real-life problems. C3(Apply)
- Apply basic data science statistical techniques by using tools on real-world datasets. C3 (Apply)

Data Warehousing & Business Intelligence — the enterprise data stack

2017 not specified

The course does not exist in HEC 2017 at either level.

2023 2-1 • prerequisite: Introduction to Data Science

Introduction to Data Warehouse and Business Intelligence; Necessities and essentials of Business Intelligence; DW Life Cycle and Basic Architecture; DW Architecture in SQL Server; Logical Model; Indexes; Physical Model; Optimizations; OLAP Operations, Queries and Query Optimization; Building the DW; Data visualization and reporting based on Datawarehouse using SSAS and Tableau; Data visualization and reporting based on Cube; Reports and Dashboard management on PowerBI; Dashboard Enrichment; Business Intelligence Tools.

Set texts: W. H. Inmon, “*Building the Data Warehouse*”, Wiley-India Edition; Ralph Kimball, “*The Data Warehouse Toolkit – Practical Techniques for Building Dimensional Data Warehouse*,” John Wiley & Sons, Inc; Matteo Golfarelli, Stefano Rizzi, “*Data Warehouse Design - Modern Principles and Methodologies*”, McGraw Hill Publisher.

2025 3+0 • no prerequisite and no outline published

- Explain the concepts, architecture, and components of Business Intelligence systems.C2 (Understand)
- Apply BI tools and techniques to extract, transform, and visualize data for decision- making.C3 (Apply)
- Design and evaluate BI solutions to support strategic and operational business decisions.. C3 (Evaluate)

Data Visualization — *exploratory graphics*

2017 not specified

The course does not exist in HEC 2017 at either level.

2023 3 (2-3) • prerequisite: Data Warehousing & Business Intelligence

Introduction of Exploratory Data Analysis and Visualization, Building Blocks and Basic Operations; Types of Exploratory Graphs, single and multi-dimensional summaries, five number summary, box plots, histogram, bar plot and others; Distributions, their representation using histograms, outliers, variance; Probability Mass Functions and their visualization; Cumulative distribution functions, percentile-based statistics, random numbers; Modelling distributions, exponential, normal, lognormal, pareto; Probability density functions, kernel density estimation; Relationship between variables, scatter plots, correlation, covariance; Estimation and Hypothesis Testing; Clustering using K-means and Hierarchical; Time series and survival analysis; Implementing concepts with R (or similar language)

Set texts: "Exploratory Data Analysis with R" by Roger D. Peng.

2025 3+0 • no prerequisite and no outline published

- Describes the basic concept and techniques for any given raw data visualization. C3 (Understand)
- Select and apply a proper data mining algorithm to discover interesting patterns C3 (Apply)
- Analyze and extract patterns to solve problems and point out how to deploy solution C4 (Analyze)
- Evaluate systematically supervised, semi-supervised, and unsupervised models and algorithms with respect to their accuracy C4 (Analyze)

Natural Language Processing — *the same title, three different courses*

2017 3(3,0) • prerequisite: none

Deterministic and stochastic grammars, Parsing algorithms, CFGs, Representing meaning / Semantics, Semantic roles, Temporal representations, Corpus-based methods, N-grams and HMMs, Smoothing and backoff, POS tagging and morphology, Information retrieval, Vector space model, Precision and recall, Information extraction, Language translation, Text classification, categorization, Bag of words model.

Set texts: Python Machine Learning, Sebastian Raschka. Publisher: Packt Publishing, 2015; Natural Language Processing with Python: Analyzing Text with the Natural Language Toolkit Latest Edition, Steven Bird, Ewan Klein and Edward Loper Publisher: O'Reilly Media, 2009; Speech and Language Processing, Latest Edition, Daniel Jurafsky and James H. Martin Publisher: Prentice Hall, 2000.

2023 3 (2-3) • prerequisite: Artificial Neural Networks & Deep Learning

Introduction & History of NLP, Parsing algorithms, Basic Text Processing, Minimum Edit Distance, Language Modeling, Spelling Correction, Text Classification, Deterministic and stochastic grammars, CFGs, Representing meaning /Semantics, Semantic roles, Semantics and Vector models, Sentiment Analysis, Temporal representations, Corpus-based methods, N-grams and HMMs, Smoothing and backoff, POS tagging and morphology, Information retrieval, Vector space model, Precision and recall, Information extraction, Relation Extraction (dependency, constituency grammar), Language translation, Text classification, categorization, Bag of words model, Question and Answering, Text Summarization

Set texts: Daniel Jurafsky and James H. Martin. 2018. Speech and Language Processing: An Introduction to Natural Language Processing. Third Edition. Prentice Hall; Foundations of Statistical Natural Language Processing, Manning and Schütze, MIT Press. Cambridge, MA: May 1999.

2025 3+0 • no prerequisite and no outline published

- Explain the fundamental concepts and techniques of Natural Language Processing, including tokenization, POS tagging, and parsing. C2 (Understand)
- Apply NLP libraries and frameworks (e.g., NLTK) to preprocess and analyze text data for real-world applications. C3 (Apply)
- Design and implement NLP-based solutions such as sentiment analysis or text classification using machine learning models. C6 (Create)

Computer Vision — images and recognition**2017 3 • prerequisite: none**

Introduction, Image formation, Spatial and frequency domain processing, Feature detection and extraction, Image registration, Segmentation, Camera calibration, Structure from motion, Motion estimation, Stereo vision, Object detection and recognition, Object tracking, 3D scene reconstruction, Context and scene understanding, Image stitching, Image-based and video-based rendering, High-performance computing paradigms for vision and image processing.,

Set texts: Computer Vision - A Modern Approach, by D. Forsyth and J. Ponce, Prentice Hall, 2003; Szeliski R., Computer Vision - Algorithms and Applications, Springer, 2011; J. R. Parker, Algorithms for Image Processing and Computer Vision, Willey Publishing Inc. 2011.

2023 3 (2-3) • prerequisite: Artificial Neural Networks & Deep Learning

Introduction to Computer Vision (Problems faced, History and Modern Advancements). Image Processing, Image filtering, Image pyramids and Fourier transform, Hough transform. Camera models, Setting up a camera model from parameters, Camera looking at a plane, Relationship of plane and horizon line, Rotation about camera center. Concatenation, Decomposition and Estimation of transformation from point correspondences, Points and planes in 2D/3D, Transformations in 2D/3D, Rotations in 2D/3D. Edge detection, corner detection. Feature descriptors and matching (HoG features, SIFT, SURF). Applications of Computer Vision Traditional Methods: Image Stitching: Making a bigger picture from smaller pictures Single View Geometry: Converting a single image into a 3D model. Applications of CV using Deep Learning: Image Detection (Localization, Historical Techniques, RCNN, FRCNN, YOLO, Retina), Image Segmentation (UNet, SegNet, MaskRCNN), Image Generation (GANN)

Set texts: Computer Vision: Algorithms and Applications, by Richard Szeliski. Reference Book:; Multiple View Geometry in Computer Vision, by Richard Hartley and Andrew Zisserman; Computer Vision: A Modern Approach, by David Forsyth and Jean Ponce.

2025 3+0 • no prerequisite and no outline published

- Understand and explain the fundamental concepts, techniques, and algorithms used in Computer Vision, including image processing, feature extraction, and object recognition. C2 (Understand)
- Implement computer vision algorithms and techniques using modern libraries and frameworks, such as OpenCV, TensorFlow, or PyTorch, to solve real-world image processing problems. C3 (Apply)
- Evaluate and assess the performance of computer vision models using metrics like accuracy, precision, recall, Intersection over Union (IoU), and mean average precision (mAP). C4 (Analyze)

Knowledge Representation & Reasoning — the symbolic half of AI**2017 not specified**

The course does not exist in HEC 2017 at either level.

2023 3 (3-0) • prerequisite: Artificial Intelligence

Propositional Logic, First-order Logic, Horn Clauses, Description Logic, Reasoning using Description Logic, Forward and Backward Chaining in Inference Engines, Semantic Networks, Ontologies and Ontology Languages, Logical Agents, Planning, Rule-based Knowledge Representation, Reasoning Under Uncertainty, Bayesian Networks Representation, Inference in Bayesian Networks, Fuzzy Logic, Inferene using Fuzzy Rules, Markov Models, Commonsense Reasoning, Explainable AI.

Set texts: Stuard Russell and Peter Norvig, Artificial Intelligence: A Modern Approach (3rd Ed.) (2015); David Poole and Alan Mackworth, Artificial Intelligence: Foundations of Computational Agents, 2nd Ed, 2017; Ronald Brachman and Hector Levesque. Knowledge Representation and Reasoning, 2004.

2025 3+0 • no prerequisite and no outline published

- Explain the fundamental concepts, models and techniques of knowledge representation and reasoning.
- Apply logical formalism, ontologies and inference mechanisms
- Use appropriate tools and methods to implement reasoning systems and evaluate their effectiveness.

Reinforcement Learning — listed in 2023, specified only in 2025**2017 not specified**

The course does not exist in HEC 2017 at either level.

2023 not specified

Listed as a BSAI domain elective but not among the 46 courses for which HEC 2023 publishes contents.

2025 3+0 • no prerequisite and no outline published

- Understand the core components of Markov decision processes and Reinforcement Learning
- Implement reinforcement learning algorithms to model and solve dynamic environments
- Evaluate reinforcement learning approaches in terms of performance scalability and applicability to real world domains

Speech Processing — listed in 2023, specified only in 2025**2017 not specified**

The course does not exist in HEC 2017 at either level.

2023 not specified

Listed as a BSAI and BSDS domain elective but not among the 46 specified courses.

2025 3+0 • no prerequisite and no outline published

- Explain fundamental concepts of speech production, perception, and signal processing, and apply them to analyze and model speech signals.
- Implement and evaluate speech processing techniques using modern programming tools and frameworks.
- Evaluate effectiveness of speech processing techniques along with ethical and societal implications of speech technologies such as privacy, surveillance, and accessibility.

Tools and Techniques in Data Science — an MS core course in 2017, an undergraduate elective in 2025**2017 not specified**

The course does not exist in HEC 2017 at either level.

2023 not specified

Not among the 46 courses for which HEC 2023 publishes contents.

2025 3+0 • no prerequisite and no outline published

- Explain the purpose and functionality of essential tools and techniques used in data science workflows. C2 (Understand)
- Apply data preprocessing, transformation, and visualization techniques using appropriate tools to solve practical problems. C3 (Apply)
- Design and implement integrated data science workflows using multiple tools for real-world datasets. C6 (Create)

Data Engineering — new in 2025**2017 not specified**

The course does not exist in HEC 2017 at either level.

2023 not specified

Not among the 46 courses for which HEC 2023 publishes contents.

2025 3+0 • no prerequisite and no outline published

- Explain the principles of data engineering, including data pipelines, ETL processes, and storage systems. C3 (Understand)
- Apply data integration and transformation techniques to build scalable and efficient data pipelines. C3 (Apply)
- Design and implement data architectures for big data and real-time analytics using modern tools and platforms. C3 (Apply)

Generative AI — new in 2025**2017 not specified**

The course does not exist in HEC 2017 at either level.

2023 not specified

Not among the 46 courses for which HEC 2023 publishes contents.

2025 3+0 • no prerequisite and no outline published

- Describe the foundational concepts, models, and architectures of Generative AI, including GANs, VAEs, and large language models. C2 (Understand)
- Implement and fine-tune generative models using modern AI frameworks such as TensorFlow or PyTorch. C3 (Apply)
- Analyze the ethical, societal, and practical implications of deploying Generative AI in real-world applications. C4 (Analyze)

Agentic AI — new in 2025**2017 not specified**

The course does not exist in HEC 2017 at either level.

2023 not specified

Not among the 46 courses for which HEC 2023 publishes contents.

2025 3+0 • no prerequisite and no outline published

- Demonstrate understanding of agentic AI concepts and design autonomous AI agents capable of reasoning, planning, and tool-use for real-world tasks.
- Implement agentic AI systems using modern frameworks with integration of APIs, databases, and external tools.
- Evaluate agentic AI systems with special focus on ethical, societal, and professional implications of deploying autonomous AI agents, and propose responsible design strategies.

Machine Learning Operations (MLOps) — *new in 2025, and where the two lanes meet***2017 not specified**

The course does not exist in HEC 2017 at either level.

2023 not specified

Not among the 46 courses for which HEC 2023 publishes contents.

2025 3+0 • no prerequisite and no outline published

- Explain concepts of MLOps and model lifecycle management.
- Apply automation for data pipelines, training, and deployment.
- Evaluate reproducibility, scalability, and governance in ML systems.

Information Retrieval — *new as a course in 2025***2017 not specified**

The course does not exist in HEC 2017 at either level.

2023 not specified

Not among the 46 courses for which HEC 2023 publishes contents.

2025 3+0 • no prerequisite and no outline published

- Explain the core principles and algorithms used in Information Retrieval systems, including indexing, ranking, and query processing.C2 (Understand)
- Apply IR techniques to build a simple search engine or retrieval system, using tools and programming languages like Python, Elasticsearch, or Apache Lucene.C3 (Apply)
- Evaluate the performance of Information Retrieval systems through metrics such as precision, recall, F1-score, and Mean Average Precision (MAP) C4 (Analyze)

Data Ethics & Security — *new in 2025***2017 not specified**

The course does not exist in HEC 2017 at either level.

2023 not specified

Not among the 46 courses for which HEC 2023 publishes contents.

2025 3+0 • no prerequisite and no outline published

- Explain the principles of data ethics, privacy laws, and governance frameworks.C2 (Understand)
- Analyze ethical and legal implications of data collection, storage, and sharing in real-world scenarios.C4 (Analyze)
- Evaluate compliance strategies for data protection and governance in organizational contexts.C5 (Evaluate)
- Design policies and guidelines to ensure ethical data usage and regulatory compliance.C6 (Create)

Optimization Techniques — new in 2025

2017 not specified

The course does not exist in HEC 2017 at either level.

2023 not specified

Not among the 46 courses for which HEC 2023 publishes contents.

2025 3+0 • no prerequisite and no outline published

- Explain the theoretical basis of optimization problems.
- Formulate and solve optimization problems using analytical and computational methods.
- Implement and evaluate most effective optimization tools and techniques for real world problems.

Stochastic Processes — in 2017 and 2025, absent from 2023

2017 3 (3,0) • prerequisite: Probability and Statistics

Discrete Markov chains, classification of states, first passage and recurrence times, absorption problems, stationary and limiting distributions. Chapman-Kolmogorov equations, Long run behavior of Markov chains, Absorption probabilities and expected times to absorption, Statistical aspects of Markov chains, The mover-stayer model, Application of a Markov chain and mover-stayer model to modeling repayment behavior of bank loans' grantees. Markov Processes in continuous time: Poisson processes, birth- death processes. Poisson process The Kolmogorov differential equations, Limiting behavior of continuous time Markov chains The Q matrix, forward and backward differential equations, imbedded Markov Chain, stationary distribution. renewal theory, Brownian Motion and its generalizations, Discrete time martingales, Conditional expectation, Definition of a martingale and examples, Optional stopping theorem, Stochastic calculus

Set texts: *Introduction to Probability Models*, 11th Ed, Sheldon M. Ross, Academic Press 2014; *Essentials of stochastic processes*, Durrett, Richard. Springer Science & Business Media, 2nd Ed, 2012; *Introduction to Stochastic Processes*, 2nd Ed, G.F. Lawler, Chapman and Hall, Probability Series, 2006.

2023 not specified

The course is absent from HEC 2023 altogether.

2025 3+0 • no prerequisite and no outline published

- Explain probability theory and stochastic models to characterize randomness and uncertainty in computational systems
- Apply probabilistic techniques and stochastic methods to analyze and model real world data and processes
- Evaluate the suitability and limitations of stochastic models in engineering and computer science applications

10.2 The infrastructure courses

Computer Networks — compulsory in every degree of every edition

2017 3+1 • prerequisite: none

Introduction and protocols architecture, basic concepts of networking, network topologies, layered architecture, physical layer functionality, data link layer functionality, multiple access techniques, circuit switching and packet switching, LAN technologies, wireless networks, MAC addressing, networking devices, network layer protocols, IPv4 and IPv6, IP addressing, subnetting, CIDR, routing protocols, transport layer protocols, ports and sockets, connection establishment, flow and congestion control, application layer protocols, latest trends in computer networks.

Set texts: Computer Networking: A Top-Down Approach Featuring the Internet, 6th edition by James F. Kurose and Keith W. Ross; Computer Networks, 5th Edition by Andrew S. Tanenbaum; Data and Computer Communications, 10th Edition by William Stallings.

2023 3 (2-3) • prerequisite: none

Introduction and protocols architecture, basic concepts of networking, network topologies, layered architecture, physical layer functionality, data link layer functionality, multiple access techniques, circuit switching and packet switching, LAN technologies, wireless networks, MAC addressing, networking devices, network layer protocols, IPv4 and IPv6, IP addressing, subnetting, CIDR, routing protocols, transport layer protocols, ports and sockets, connection establishment, flow and congestion control, application layer protocols, latest trends in computer networks.

2025 3 • no prerequisite and no outline published

- Explain the fundamental concepts of computer networking, including protocols, topologies, and models (OSI, TCP/IP).
- Configure and troubleshoot basic network devices and connections.
- Analyze network security threats and mitigation techniques.
- Demonstrate understanding of data transmission and error handling.
- Design simple network architectures to meet organizational needs.

System & Network Administration — compulsory, compulsory, then optional

2017 3 (3,0) • prerequisite: Operating System

Introduction To System Administration. SA Components. Server Environment (Microsoft and Linux). Reliable Products, Server Hardware Costing, Maintenance Contracts and Spare Parts, Maintaining Data Integrity, Client Server OS Configuration, Providing Remote Console Access. Comparative Analysis of OS: Important Attributes, Key Features, Pros and Cons. Linux Installation and Verification, Configuring Local Services and Managing Basic System Issues. Administer Users and Groups. Software Management. Managing Network Services and Network Monitoring Tools. Boot Management and Process Management. IP Tables and Filtering. Securing Network Traffic. Advanced File Systems and Logs. Bash Shell Scripting. Configuring Servers (FTP, NFS, Samba, DHCP, DNS and Apache).

Set texts: The Practice of System and Network Administration, Second Edition by Thomas Limoncelli, Christina Hogan and Strata Chalup, Addison-Wesley Professional; 2nd Edition (2007). ISBN-10: 0321492668; Red Hat Enterprise Linux 6 Bible: Administering Enterprise Linux Systems by William vonHagen, 2011; Studyguide for Practice of System and Network Administration by Thomas A. Limoncelli, Cram101; 2nd Edition (2011). ISBN-10: 1428851755.

2023 not specified

A BSIT domain core course in Semester 5, and unspecified for the same reason: HEC 2023 has no Information Technology course-contents table.

2025 2+1 • no prerequisite and no outline published

- Explain principles of system and network administration.
- Configure and troubleshoot network services and operating systems.
- Use administrative tools to monitor and secure systems.
- Demonstrate professionalism in managing shared computing resources.
- Work effectively in teams to maintain IT infrastructure.

Cloud Computing — *unspecified, unspecified, then compulsory***2017 not specified**

Named once, in the BS Software Engineering elective list, at 3-0. It is absent from the 75-course bachelor list and no specification exists anywhere in the booklet.

2023 not specified

Listed as a BSIT domain elective and in the master course list, but HEC 2023 publishes contents only for courses common to all computing degrees and for the AI, Data Science and Cyber Security clusters. There is no Information Technology table.

2025 3 • no prerequisite and no outline published

- Describe fundamental cloud computing models and services (IaaS, PaaS, SaaS).
- Deploy and manage applications in cloud environments.
- Analyze the benefits and challenges of cloud computing.
- Implement basic cloud security and compliance measures.
- Evaluate cloud solutions for scalability, cost, and performance.

Cloud Infrastructure and Services — *the 2025 IT-cluster near-duplicate of Cloud Computing***2017 not specified**

The course does not exist in HEC 2017 at either level.

2023 not specified

Not among the 46 courses for which HEC 2023 publishes contents.

2025 2+1 • no prerequisite and no outline published

- Explain cloud service models and infrastructure components.
- Deploy and manage virtualized cloud environments.
- Use cloud management tools for provisioning and monitoring.
- Evaluate legal and compliance issues in cloud computing.
- Engage in lifelong learning to adapt to cloud innovations.

Virtual Systems and Services — 2017's nearest thing to a cloud course

2017 3 • prerequisite: Programming Fundamentals

This course will investigate the current state of virtualization in computing systems. Virtualization at both the hardware and software levels will be examined, with emphasis on the hypervisor configurations of systems such as VMware, Zen and Hyper-V. The features and limitations of virtual environments will be considered, along with several case studies used to demonstrate the configuration and management of such systems. Para-virtualized software components will be analyzed and their pros and cons discussed. Processor and peripheral support for virtualization will also be examined, with a focus on emerging hardware features and the future of virtualization.

Set texts: Handbook of Virtual Environments: Design, Implementation, and Applications (Human Factors and Ergonomics), Edited by Kay M Stanney, Lawrence Erlbaum Associates Virtual Reality Technology by GRIGORE.

2023 not specified

A BSIT domain elective example, unspecified: HEC 2023 has no Information Technology course-contents table.

2025 not specified

The course is gone. Its ground is covered by Cloud Computing and, in the infrastructure cluster, by Microservices Architecture and Docker Containers.

Information Technology Infrastructure — the architecture-side companion to administration

2017 3 (3,0) • prerequisite: none

Definition of IT Infrastructure, Non-functional Attributes, Availability Concepts, Sources of Unavailability, Availability Patterns. Performance. Security Concepts. Data centres. Servers: Availability, Performance, Security. Networking: Building Blocks, Availability, Performance, Security. Storage: Availability, Performance, Security. Virtualization: Availability, Performance, Security. Operating Systems: Building Blocks, Implementing Various OSs, OS availability, OS Performance, OS Security. End User Devices: Building Blocks, Device Availability, Performance, Security. IT Infrastructure Management. Service Delivery Processes. Service Support Processes. Ethics, Trends, organizational and technical issues related to IT infrastructure.

Set texts: IT Infrastructure Architecture: Infrastructure building blocks and concepts by Sjaak Laan, Lulu.com (November 5, 2011). ISBN-10: 1447881281; IT Infrastructure and its Management by Prof Phalguni Gupta, Tata McGraw Hill Education Private Limited (October 6, 2009). ISBN-10: 0070699798; IT Architecture for Dummies by Kalani Kirk Hausman and Susan Cook, For Dummies; 1st Edition (November 9, 2010). ISBN-10: 0470554231.

2023 not specified

A BSIT domain core course, unspecified for the same reason.

2025 2+1 • no prerequisite and no outline published

- Identify components of IT infrastructure and their roles.
- Design basic infrastructure models for organizational needs.
- Apply tools for infrastructure monitoring and support.
- Assess societal and organizational impacts of infrastructure decisions.
- Engage in self-directed learning to stay updated with infrastructure trends.

Network Security — specified in 2023 only because it is common to every degree

2017 not specified

The course does not exist in HEC 2017 at either level.

2023 3 (2-3) • prerequisite: Introduction to Cyber Security

Introduction to network security, Networking Concepts and Protocols, Network Threats and Vulnerabilities, Network Security Planning and Policy, Access Control, Defense against Network Attacks, DOS and DDOS detection and prevention, Firewalls, Intrusion Detection and Prevention Systems, Antivirus Filtering, Naming and DNS Security, DNSSEC, IP security, Secure Sockets Layer, VPN, Packet Sniffing and spoofing, Honeypot, Ethernet Security, Wireless Security, Wireless Attacks, Wireless LAN Security with 802.11i, Wireless Security Protocols, Wireless Intrusion Detection, Physical access and Security, Tor Network, Network Forensics. Defense against Network Attacks.

Set texts: Network Security Assessment: Know Your Network by Chris McNab, 3rd Edition or latest; Corporate Computer Security, by Randall J. Boyle, 3th Edition; Bulletproof Wireless Security by Praphul Chandra.

2025 2+1 • no prerequisite and no outline published

- Explain common network threats and vulnerabilities. (BT2 – Understand) (PLO1)
- Configure basic network security mechanisms such as firewalls and VPNs. (BT3 – Apply) (PLO5)
- Design and develop network defense solutions (BT3-Apply)(PLO3)
- Evaluate existing security architectures to identify strengths and weaknesses. (BT5 – Evaluate) (PLO1)

Parallel and Distributed Computing — a domain core in BSAI, BSDS and BSIT alike**2017 not specified**

The course does not exist in HEC 2017 at either level.

2023 3 (2-3) • prerequisite: Object Oriented Programming, Operating Systems

Asynchronous/synchronous computation/communication, concurrency control, fault tolerance, GPU architecture and programming, heterogeneity, interconnection topologies, load balancing, memory consistency model, memory hierarchies, Message passing interface (MPI), MIMD/SIMD, multithreaded programming, parallel algorithms & architectures, parallel I/O, performance analysis and tuning, power, programming models (data parallel, task parallel, process-centric, shared/distributed memory), scalability and performance studies, scheduling, storage systems, synchronization, and tools (Cuda, Swift, Globus, Condor, Amazon AWS, OpenStack, Cilk, gdb, threads, MPICH, OpenMP, Hadoop, FUSE).

Set texts: Distributed Systems: Principles and Paradigms, A. S. Tanenbaum and M. V. Steen, Prentice Hall, 2nd Edition, 2007; Distributed and Cloud Computing: Clusters, Grids, Clouds, and the Future Internet, K Hwang, J Dongarra and GC. C. Fox, Elsevier, 1st Ed. 65 Course Contents of Mathematics & Supporting Courses.

2025 3 • no prerequisite and no outline published

- Explain concepts of concurrency, synchronization, and distributed architectures. (C2: Understand)
- Apply parallel programming libraries to solve computational problems. (C3: Apply)
- Analyze the performance trade-offs between sequential and parallel solutions. (C4: Analyze)

10.3 The networking depth courses

Every one of these is new in HEC 2025. Neither 2017 nor 2023 contains a course beyond *Computer Networks* and *Network Security*, which is why Part 7 concludes that a CCNA-aligned track is only buildable under the 2025 curriculum.

Data Communication — the physical and link layer, split out from Computer Networks**2017 not specified**

The course does not exist in HEC 2017 at either level.

2023 not specified

Not among the 46 courses for which HEC 2023 publishes contents.

2025 2+1 • no prerequisite and no outline published

- Explain data transmission methods and communication protocols.
- Analyze performance of communication systems.
- Use simulation tools to model data communication networks.
- Collaborate in teams to design communication solutions.
- Present technical findings in written and oral formats.

Internetworking — routers, switches and gateways — the nearest thing to routing and switching
2017 not specified

The course does not exist in HEC 2017 at either level.

2023 not specified

Not among the 46 courses for which HEC 2023 publishes contents.

2025 (2+1)/3 • no prerequisite and no outline published

- Identify the role of routers, switches, and gateways in internetwork communication.
- Configure internetworking devices for data transmission across heterogeneous networks.
- Evaluate performance trade-offs in different internetworking topologies.

Network Management — monitoring, faults and metrics
2017 not specified

The course does not exist in HEC 2017 at either level.

2023 not specified

Not among the 46 courses for which HEC 2023 publishes contents.

2025 (2+1)/3 • no prerequisite and no outline published

- Explain standard tools and frameworks for managing network infrastructure.
- Apply monitoring techniques to track network performance and faults.
- Analyze collected metrics to diagnose and resolve management issues.

Network Performance — throughput, latency, jitter and QoS
2017 not specified

The course does not exist in HEC 2017 at either level.

2023 not specified

Not among the 46 courses for which HEC 2023 publishes contents.

2025 (2+1)/3 • no prerequisite and no outline published

- Explain key concepts, metrics, and models related to network performance (e.g., throughput, latency, jitter, QoS).
- Apply analytical and simulation techniques to evaluate the performance of computer networks.
- Analyze the impact of protocols, architectures, and traffic patterns on network efficiency.
- Design and propose solutions to enhance performance in real-world networking scenarios.

Wireless and Mobile Networks — *the wireless half of the CCNA blueprint*

2017 not specified

The course does not exist in HEC 2017 at either level.

2023 not specified

Not among the 46 courses for which HEC 2023 publishes contents.

2025 2+1 • no prerequisite and no outline published

- Describe wireless technologies and mobile communication standards.
- Analyze challenges in wireless network design and deployment.
- Apply tools to configure and test wireless networks.
- Assess health and safety concerns in mobile network environments.
- Demonstrate effective teamwork in wireless network projects.

Software Defined Networks (SDN) — *controller-based networking*

2017 not specified

The course does not exist in HEC 2017 at either level.

2023 not specified

Not among the 46 courses for which HEC 2023 publishes contents.

2025 (2+1)/3 • no prerequisite and no outline published

- Describe the architecture and principles of SDN.
- Configure SDN controllers to manage dynamic traffic flows.
- Design SDN solutions to optimize scalability and security in enterprise environments.

Network Programmability and Automation — *APIs, scripts and playbooks*

2017 not specified

The course does not exist in HEC 2017 at either level.

2023 not specified

Not among the 46 courses for which HEC 2023 publishes contents.

2025 (2+1)/3 • no prerequisite and no outline published

- Demonstrate use of APIs for programmable network configurations.
- Develop scripts and playbooks for automated deployment of network services.
- Evaluate trade-offs between manual and automated approaches in network management.

Network Automation and Orchestration — *provisioning and multi-domain orchestration*

2017 not specified

The course does not exist in HEC 2017 at either level.

2023 not specified

Not among the 46 courses for which HEC 2023 publishes contents.

2025 (2+1)/3 • no prerequisite and no outline published

- Demonstrate automation workflows for network provisioning.
- Implement orchestration tools to coordinate multi-domain services
- Design end-to-end automated orchestration solutions.

Next-Generation Networks and Edge Computing — 5G and edge**2017 not specified**

The course does not exist in HEC 2017 at either level.

2023 not specified

Not among the 46 courses for which HEC 2023 publishes contents.

2025 3+0 • no prerequisite and no outline published

- Explain the concepts of edge computing and 5G architectures.
- Analyze performance benefits and latency reductions enabled by edge computing.
- Design use-case scenarios integrating 5G with edge infrastructure.

Enterprise Network Infrastructure Engineering Design — the design capstone of the infrastructure cluster**2017 not specified**

The course does not exist in HEC 2017 at either level.

2023 not specified

Not among the 46 courses for which HEC 2023 publishes contents.

2025 (2+1)/3 • no prerequisite and no outline published

- Apply systematic design methodologies for enterprise network infrastructures.
- Evaluate scalability, availability, and resilience requirements in enterprise networks.
- Create a comprehensive enterprise network design with redundancy and security features.

i Reading these three columns as a drafting instruction

For any course in this part the practical rule is the same. If HEC 2023 gives an outline, use it — it is the most recent detailed specification and it carries prerequisites. If only HEC 2017 gives one, use that and modernise the tooling; the structure is almost always still right, and it is the only detailed text that exists for the Information Technology courses. If only HEC 2025 gives outcomes, the department is writing the course from scratch and should say so in its own course file, mapping each locally written topic to one of the published CLOs so that accreditation has something to check against.

Sources

Every statement about the curriculum is drawn from the three booklets below. Page references are to the PDF page, not the printed page number. Part 7 additionally draws on Cisco's own published blueprint, cited separately after the table.

Edition	Document and the sections used
2017	<i>Curriculum of BS & MS (CS, SE & IT)</i> , HEC, 171 pp. BS-IT curriculum structure and study plan, pp. 27–33; BS-CS domain courses, pp. 23–24; BS-SE elective list (the only mention of Cloud Computing), p. 38; the 75-course bachelor list, pp. 69–70; course details for Artificial Intelligence, Big Data Analytics, Computer Networks, Computer Vision, Natural Language Processing, System and Network Administration, IT Infrastructure and Virtual Systems and Services, pp. 71–130; MS Data Science programme, pp. 47–50
2023	<i>BS Curriculum, Computing Disciplines</i> , HEC / NCEAC, 100 pp. Generic computing structure, pp. 10–12; BSCS model and study plan, pp. 17–19; BSAI, pp. 23–25; BSDS, pp. 26–28; BSIT model and study plan, pp. 41–43; full list of BS courses, pp. 50–53; course contents, pp. 54–100 — note the scope statement at the head of section 7, which limits published contents to the courses common to all computing degrees plus the AI, Data Science and Cyber Security cluster tables (7.4–7.8)
2025	<i>HEC Curriculum of Computer Science</i> , HEC, 86 pp. Programme structure and interdisciplinary courses, pp. 15–16; scheme of studies for Semesters I–VIII, pp. 17–20; specializations and the guidelines for them, pp. 20–21; Data Science cluster, pp. 21–22; Artificial Intelligence cluster, p. 22; Information Technology cluster, p. 23; Network Infrastructure & Cloud Computing cluster, pp. 17–18 of the specialization listing; course learning outcomes for the mandatory major courses, pp. 24–26, and for the specialization courses, pp. 37–64; AI specialization MOOC and certification recommendations, pp. 67–73

i The CCNA material in Part 7

The blueprint in Part 7 is Cisco's, not HEC's. Part 7 is aligned throughout to **CCNA 200-301 version 2.0**, taken from Cisco's own published exam-topics document, *Implementing and Administering Cisco Solutions (200-301 CCNA) v2.0*, marked Cisco Public and dated 2026. The five domains, their weights and the twenty-nine numbered topics are as that document gives them. The verb counts in Part 7.2 — eight topics beginning *Troubleshoot* or *Diagnose*, eight beginning *Configure*, six *Describe* — were counted from it directly rather than taken from any secondary source.

Version 2.0 was announced on 20 May 2026, replaces v1.1 on 3 February 2027 and keeps the same exam number. The v1.1 comparison in Part 7.5 draws on Cisco's v1.1 exam-topics document and on published analyses of the transition.

Two cautions. Cisco revises blueprints without renaming the exam, so **verify against [cisco.com/go/certroadmap](https://www.cisco.com/go/certroadmap) before committing a scheme of studies** — the version number above is the check to run, not an answer to trust indefinitely. And the mapping in Part 7.3 is a judgement made by comparing published HEC course outlines and learning outcomes against published CCNA topic statements; it is a planning aid, not an accreditation of equivalence, and no HEC course has been certified by Cisco as covering any part of the exam.

i On the two courses this document does not settle

The 2023 booklet lists both *Data Science* and *Introduction to Data Science* in its master course list, and gives a full specification only for the second. Likewise it lists both *Artificial Neural*

Networks & Deep Learning and *Deep Learning* with overlapping outlines. Where the two conflict this document follows the specification with the published outline and prerequisites, and says so in the relevant card.